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Yoshioka et al.

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(54) **TERMINAL AND COMMUNICATION METHOD**

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CPC **H04W 72/0446** (2013.01)

(58) **Field of Classification Search**
CPC **H04W 72/0446**
See application file for complete search history.

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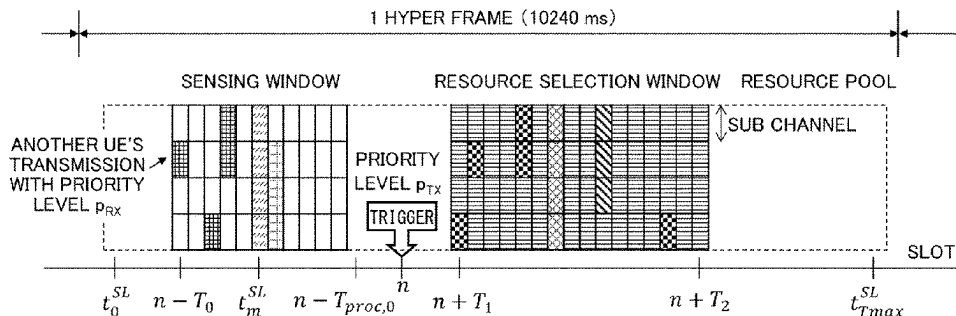
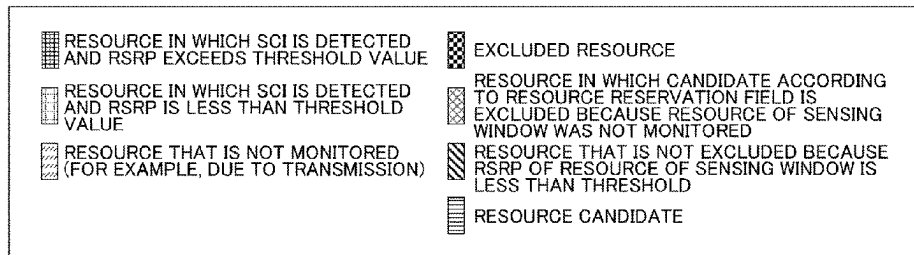
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(57) **ABSTRACT**

A terminal including a receiving unit configured to perform sensing of resource reservation information and resource allocation information transmitted from another terminal by using a sensing window; a control unit configured to determine an available resource set, by identifying a predetermined resource in a resource selection window, based on the resource reservation information and the resource allocation information; and a transmitting unit configured to execute transmission to the another terminal by using a resource selected from the resource set.

7 Claims, 16 Drawing Sheets



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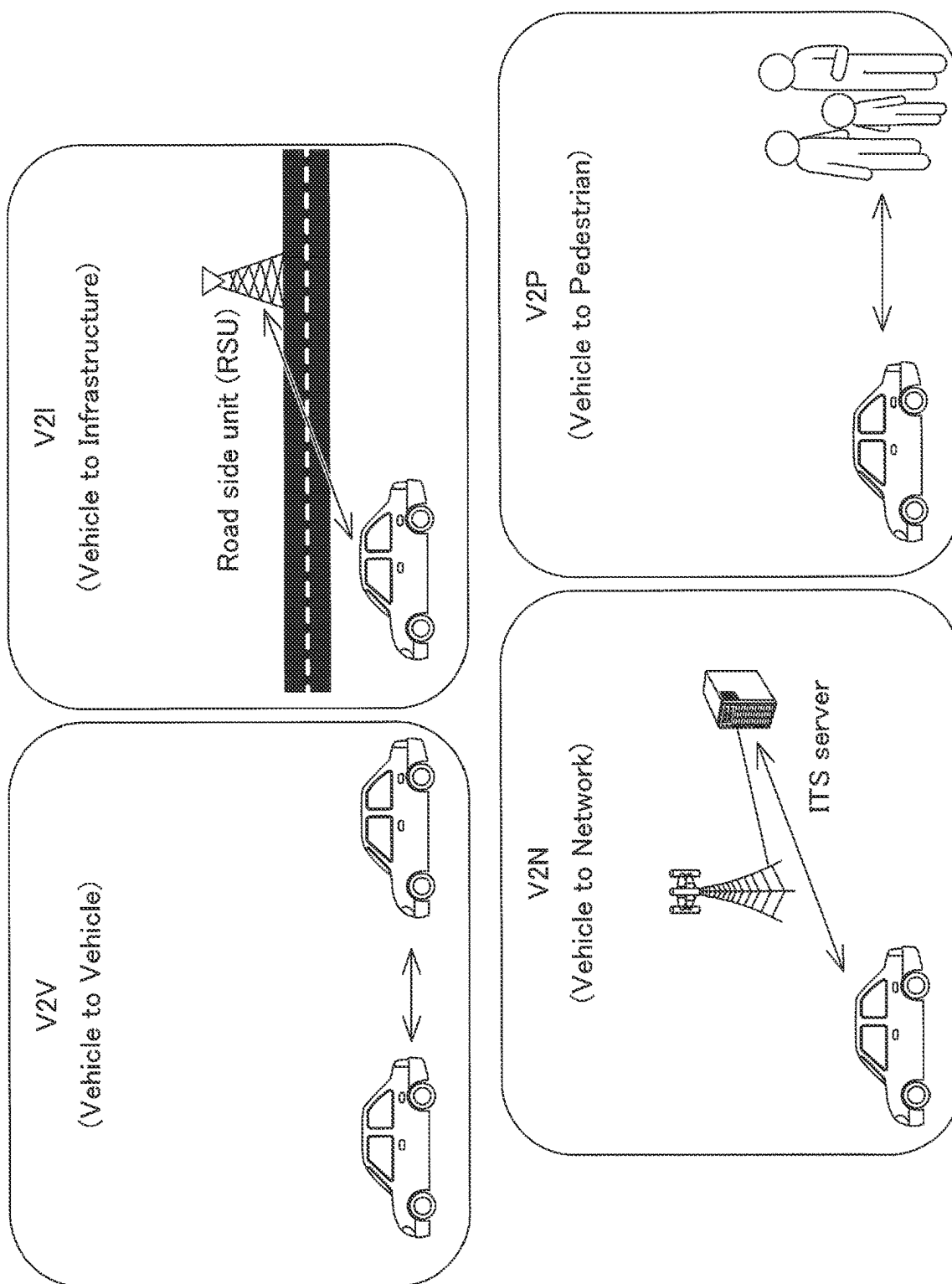


FIG.1

FIG.2

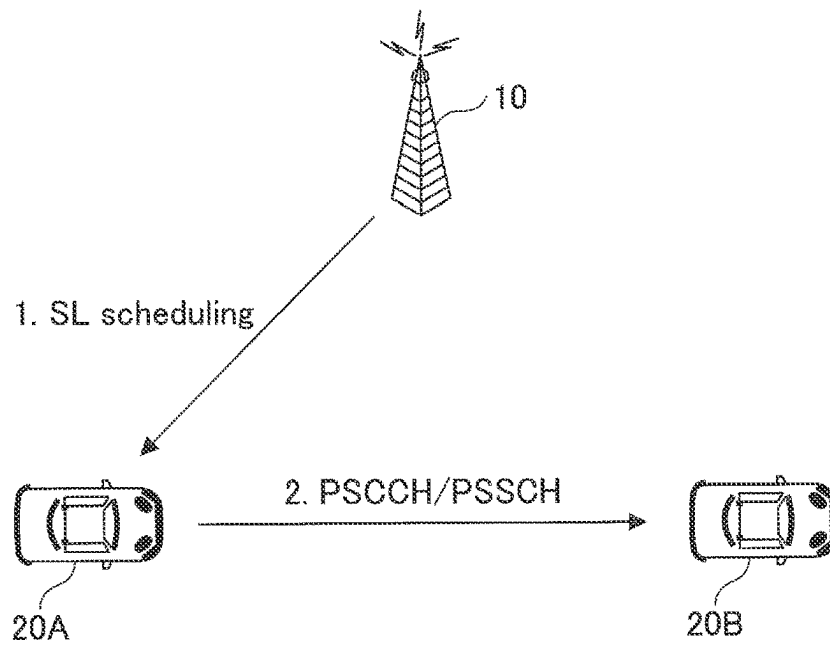


FIG.3

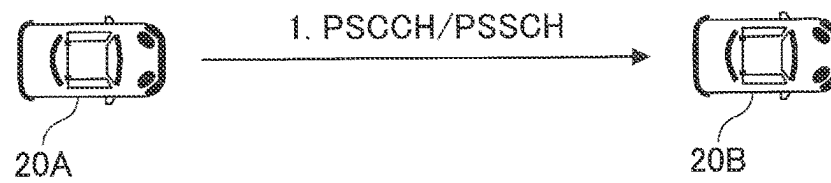


FIG.4

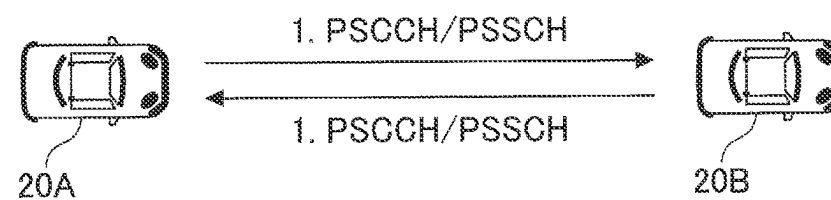


FIG.5

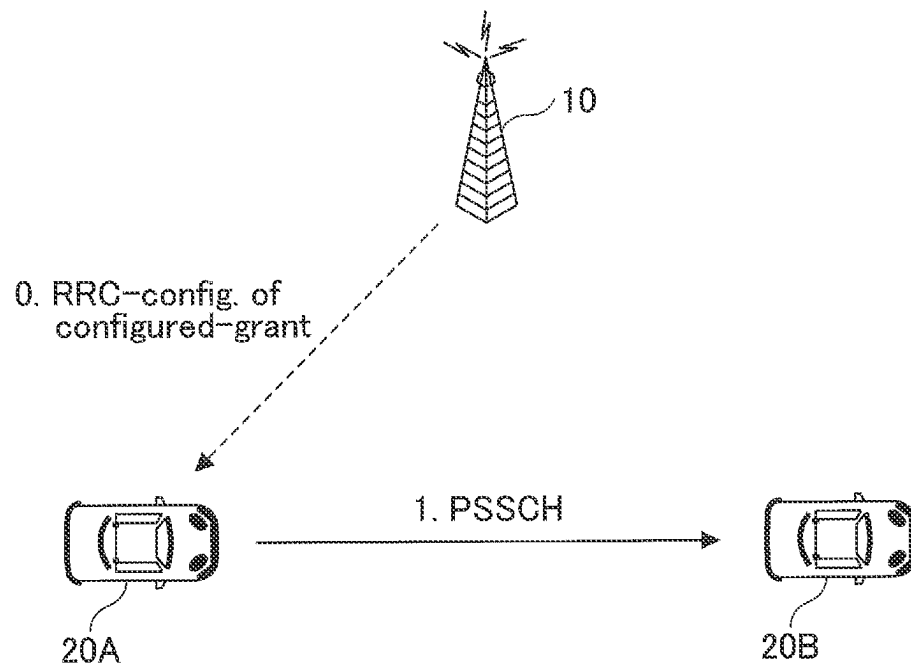


FIG.6

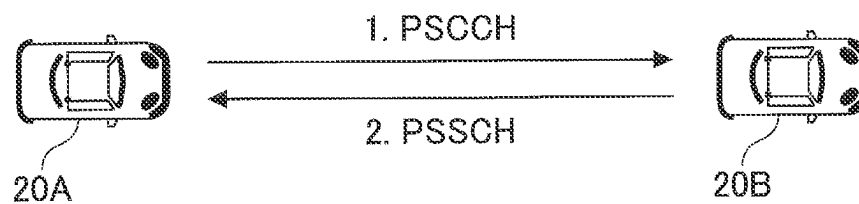


FIG. 7

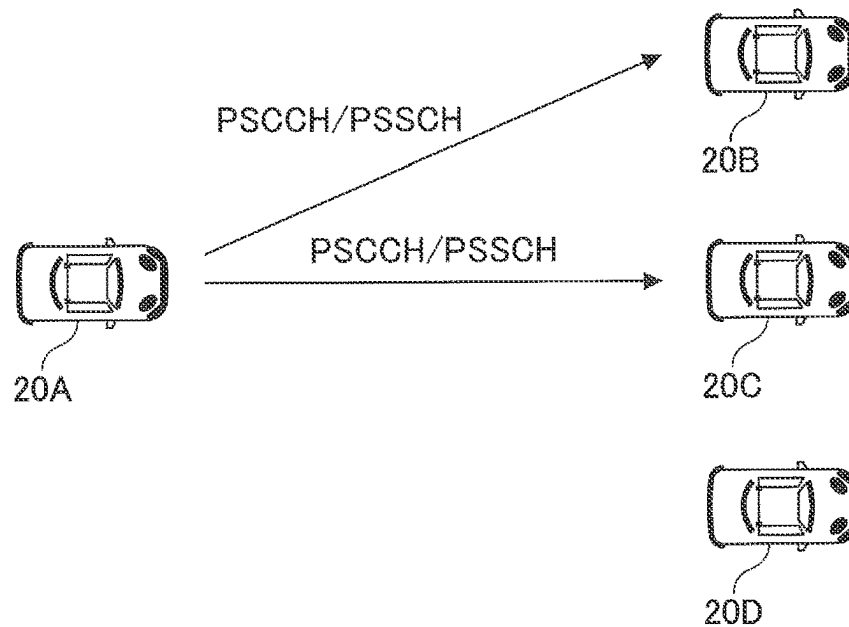


FIG. 8

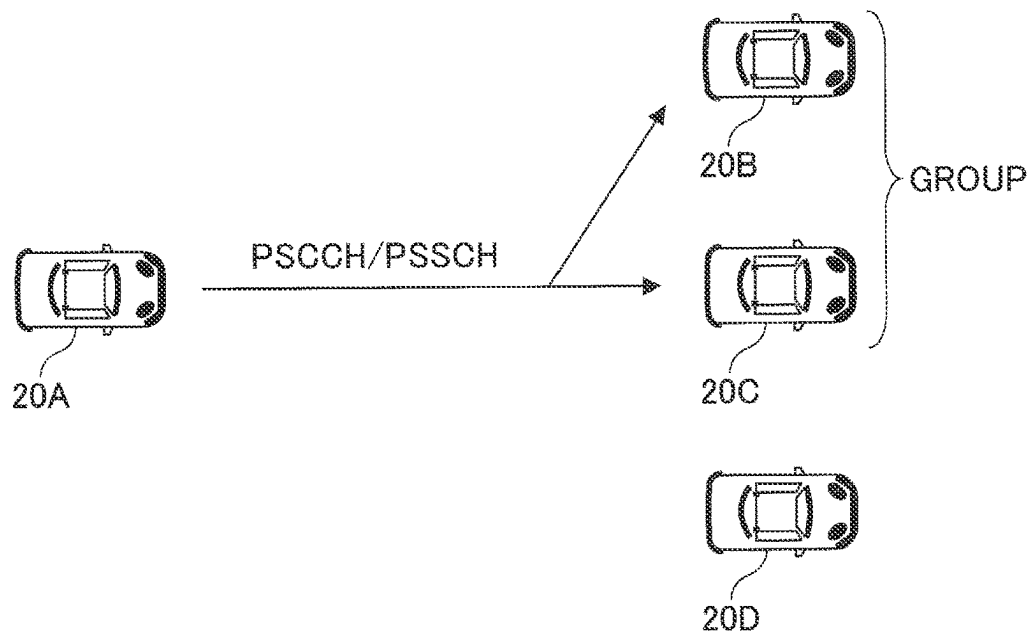


FIG.9

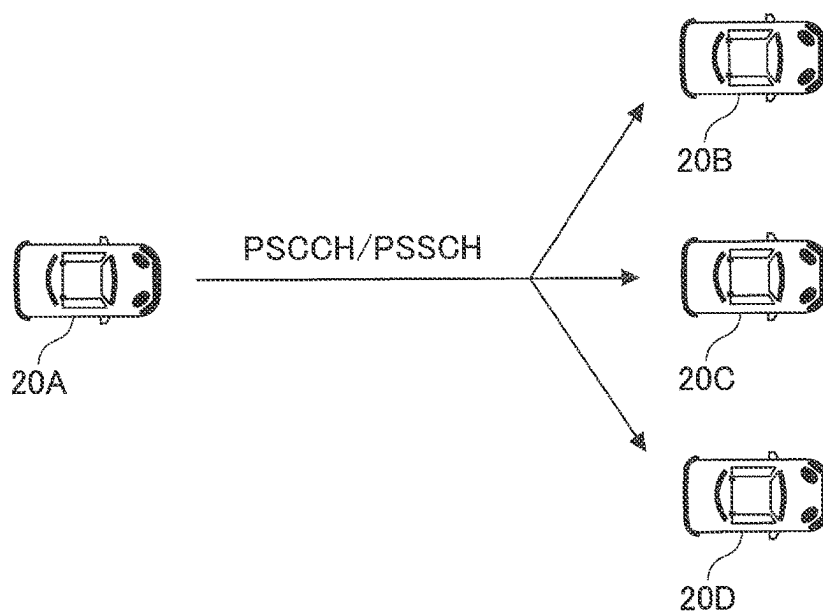


FIG.10

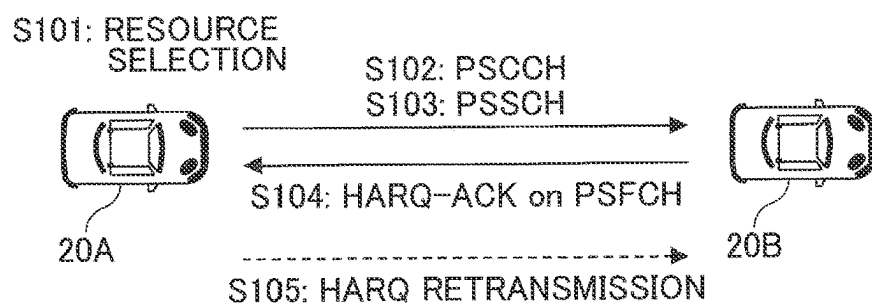


FIG.11

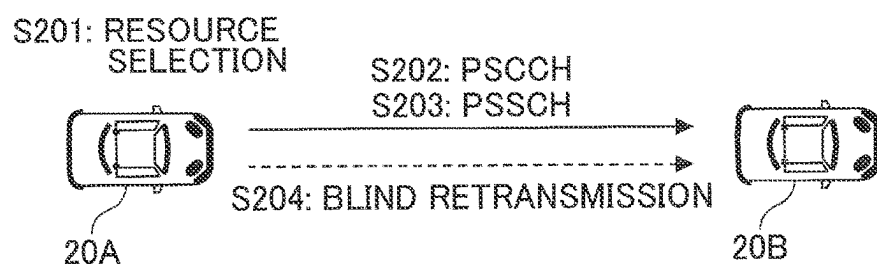
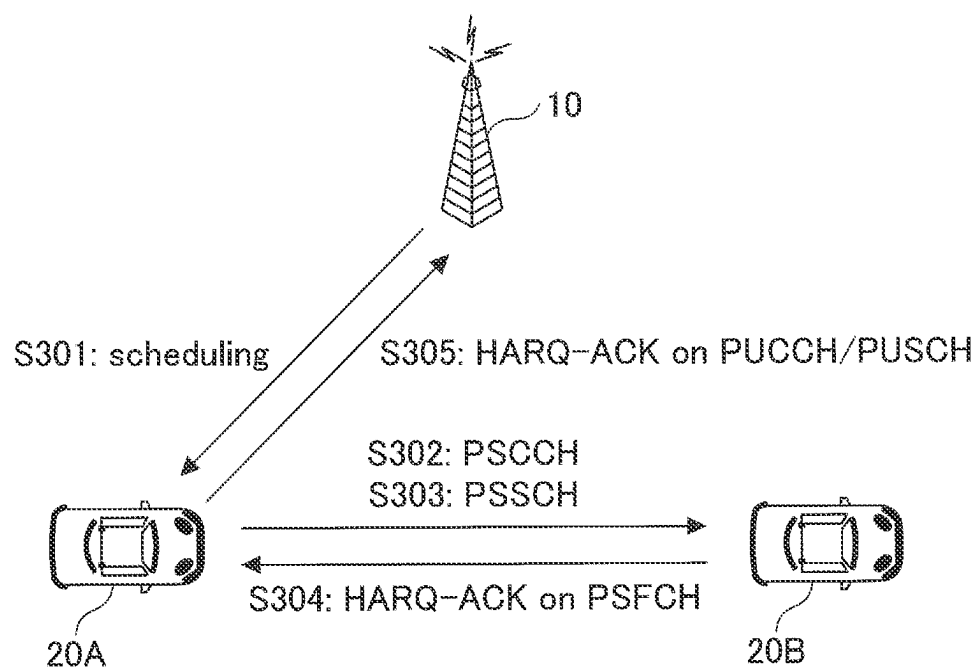


FIG. 12



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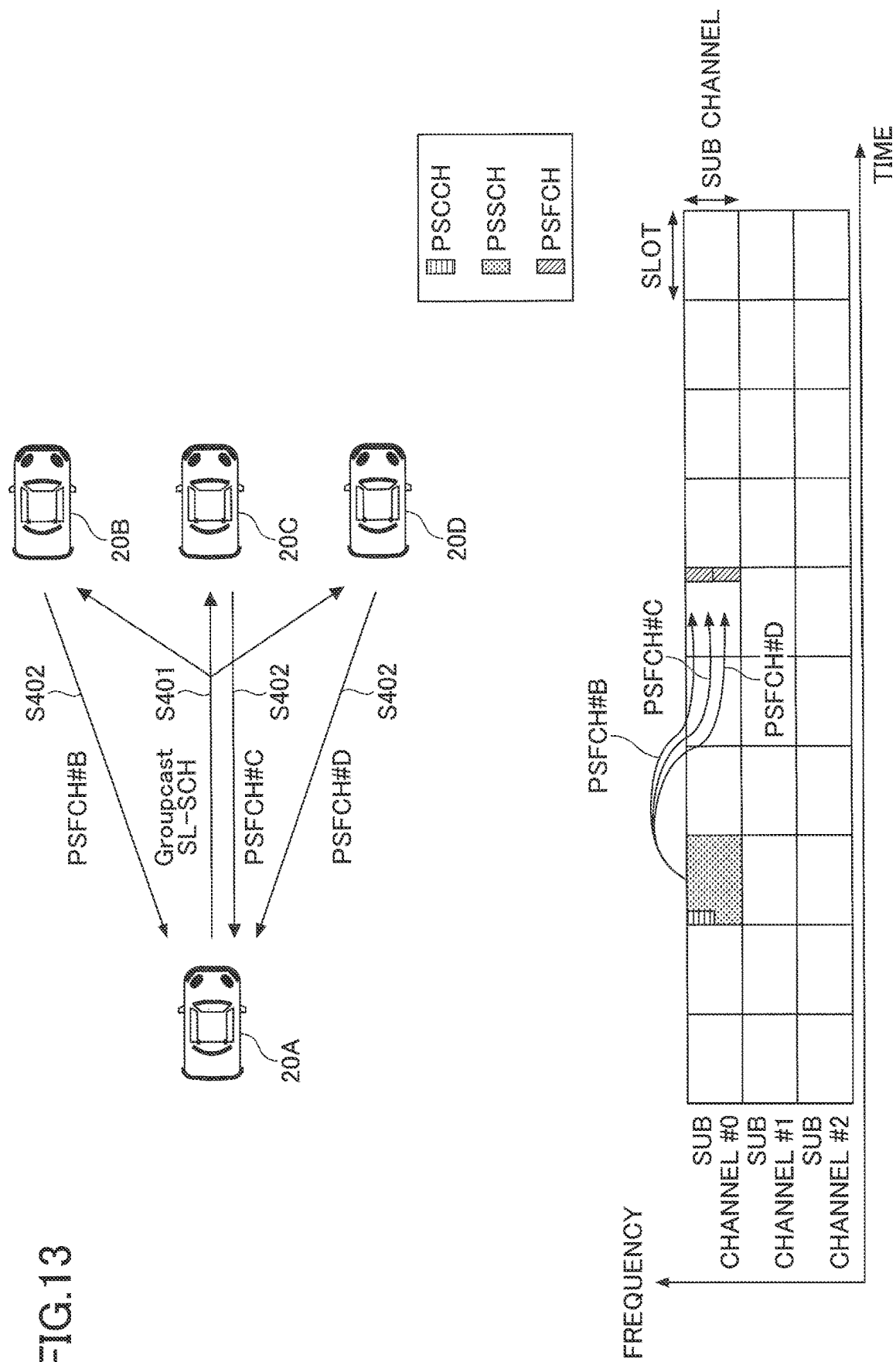
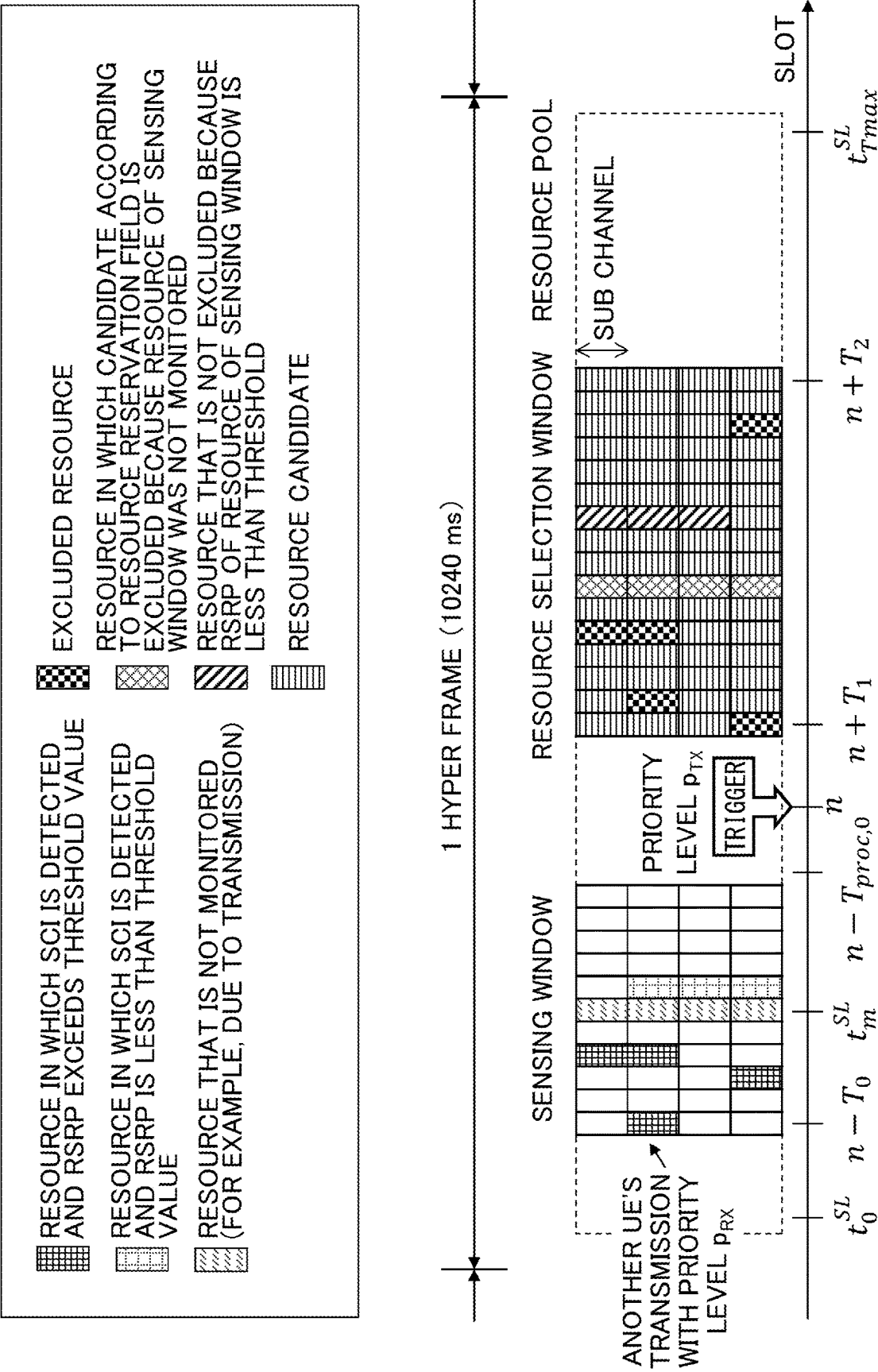
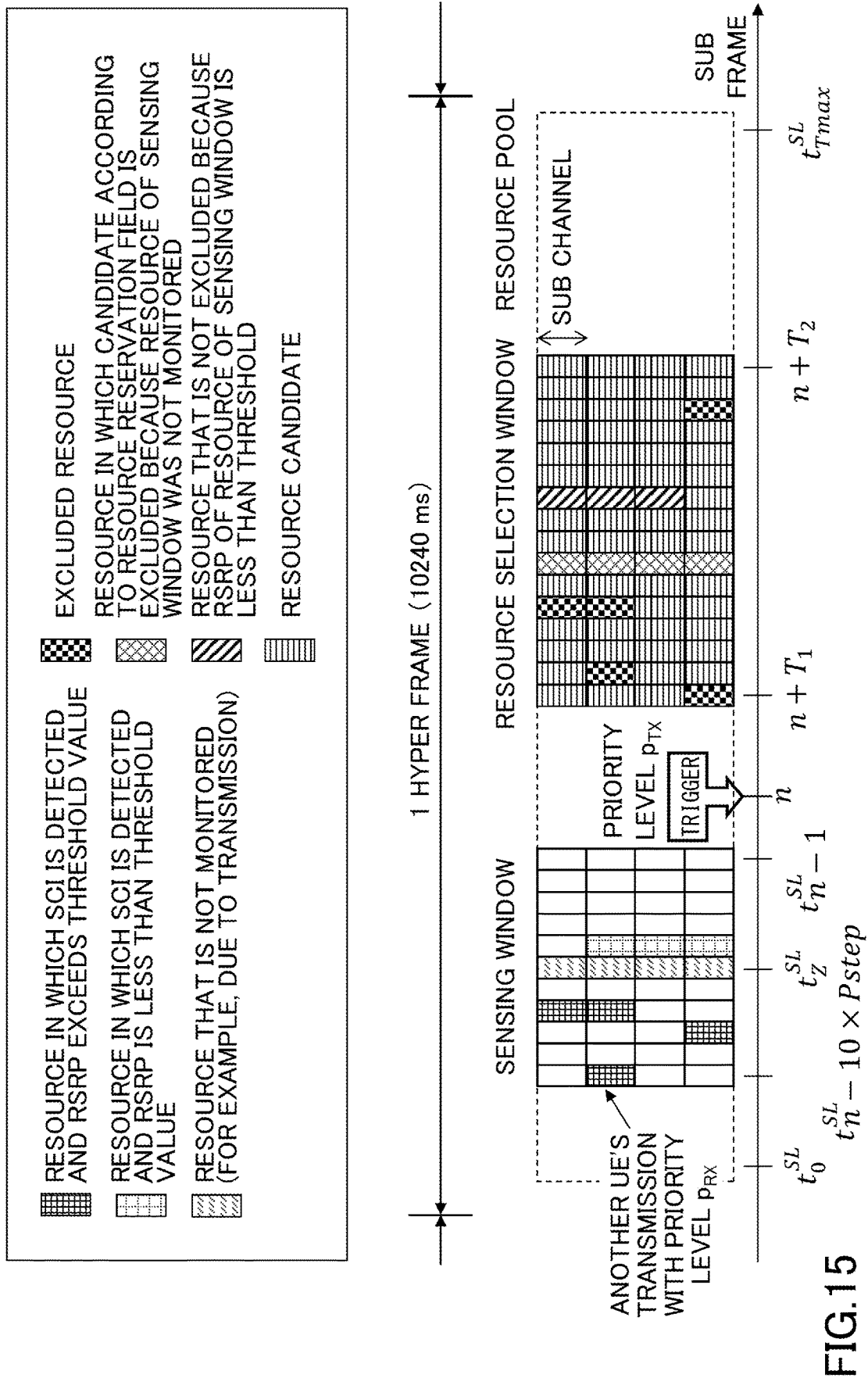


FIG.14





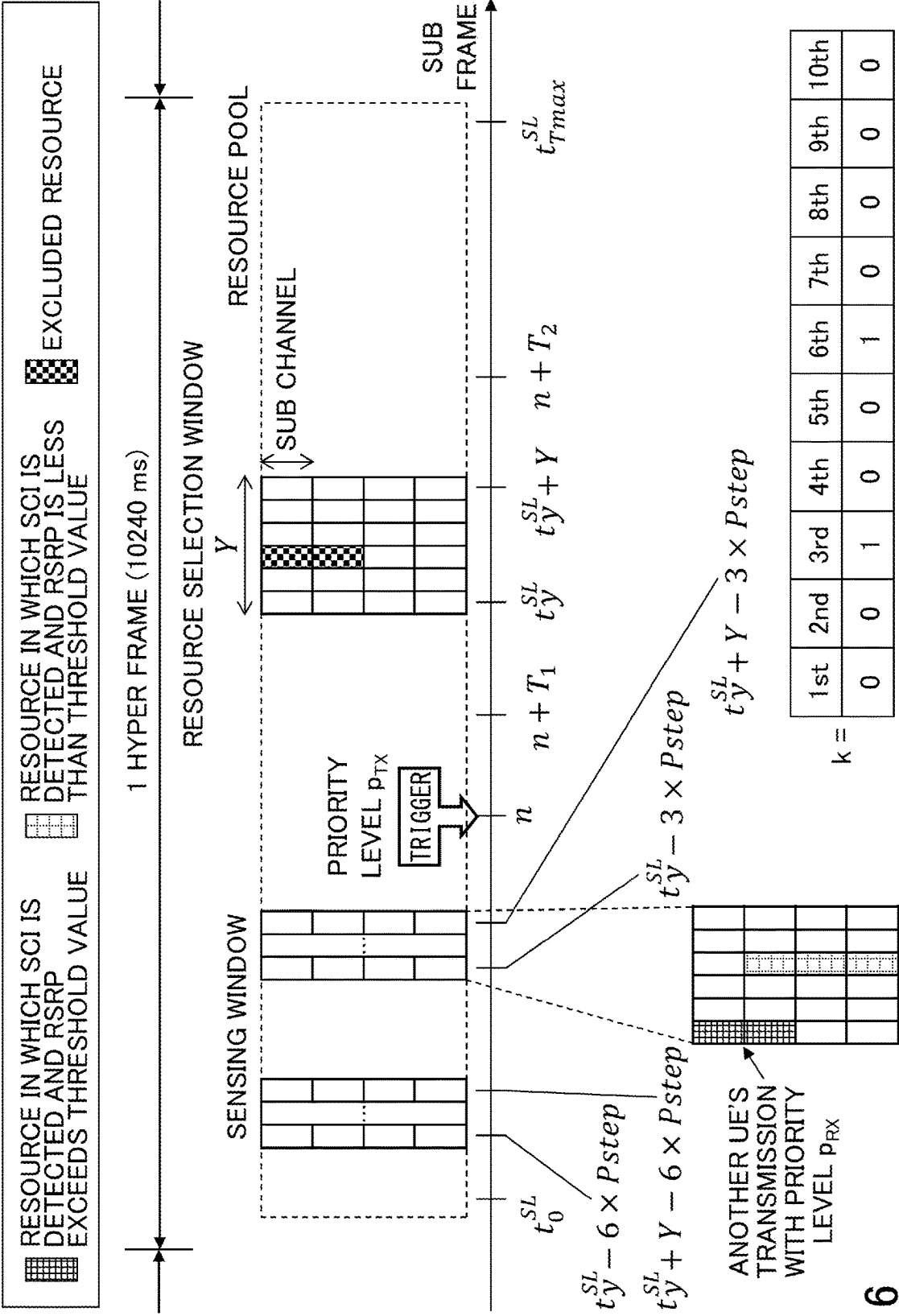
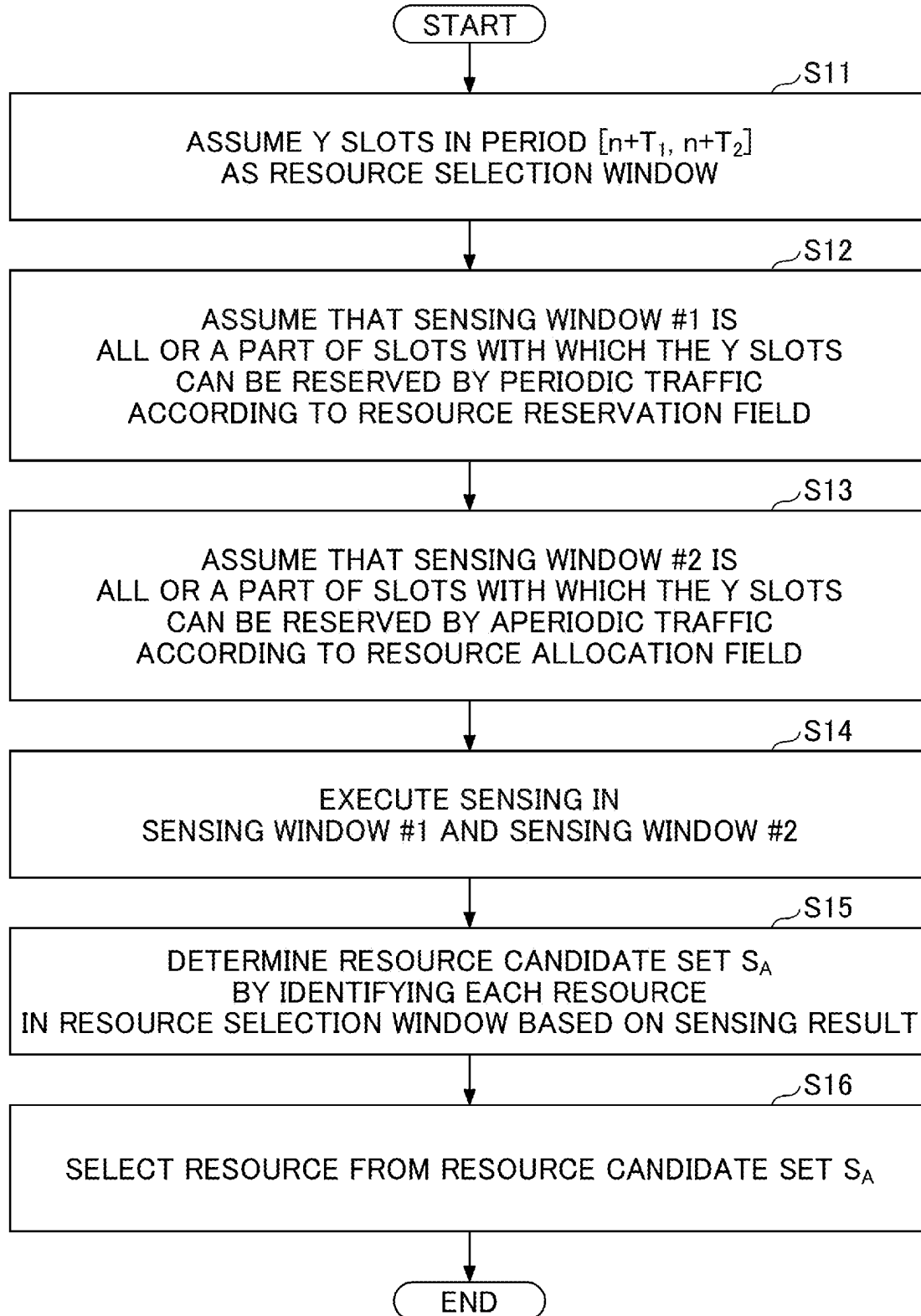


FIG.16

FIG.17



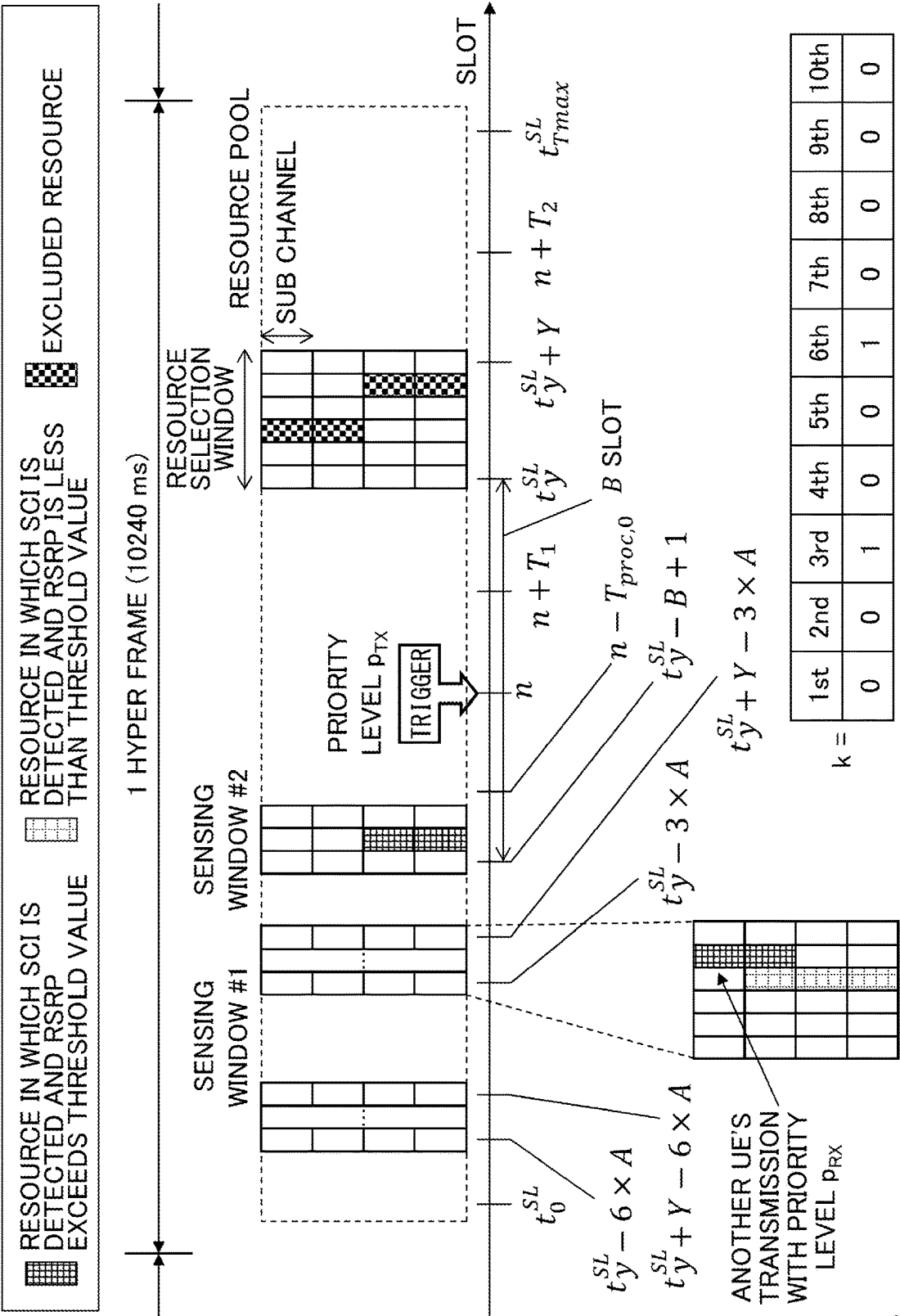
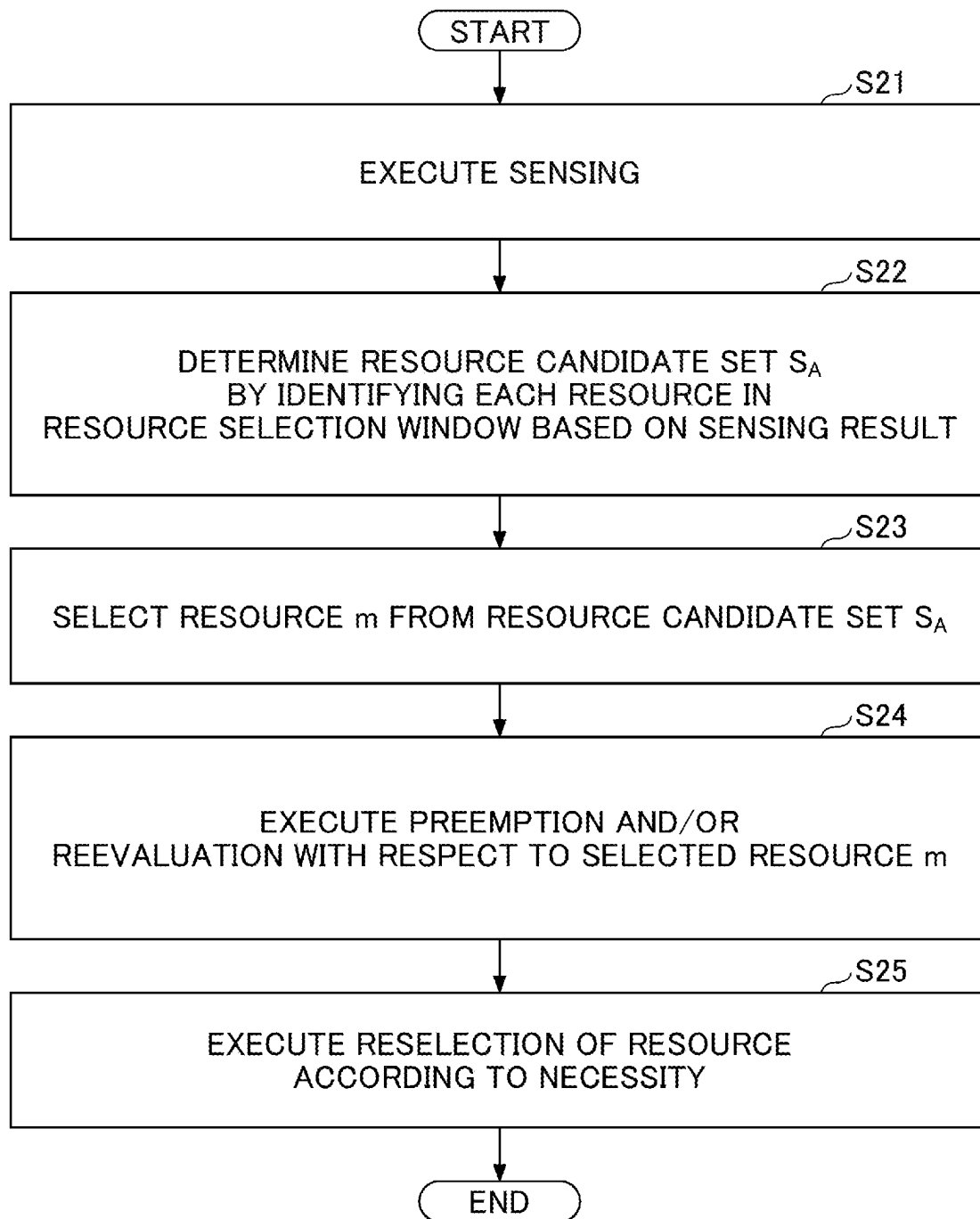


FIG.18

FIG.19



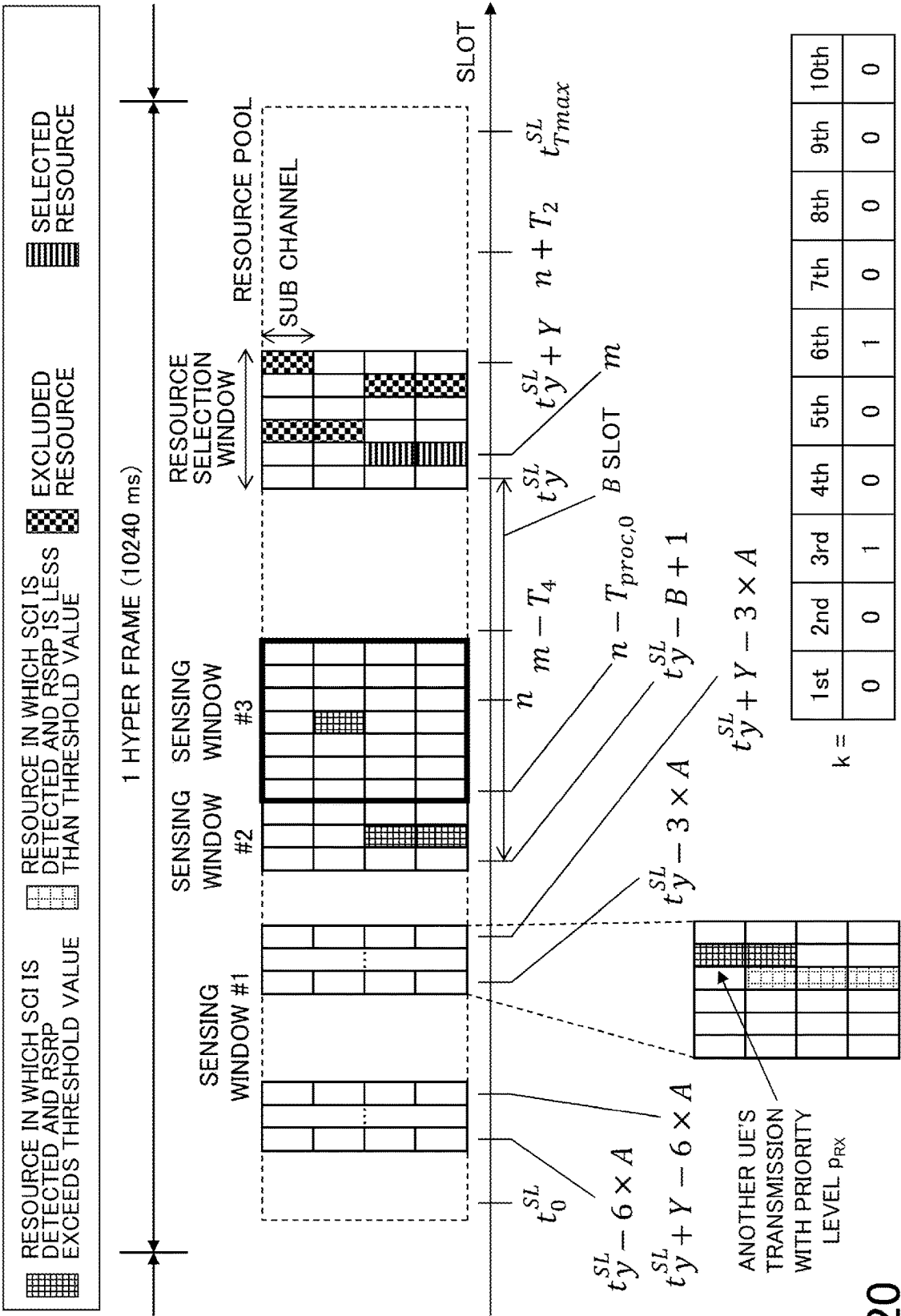


FIG.20

FIG.21

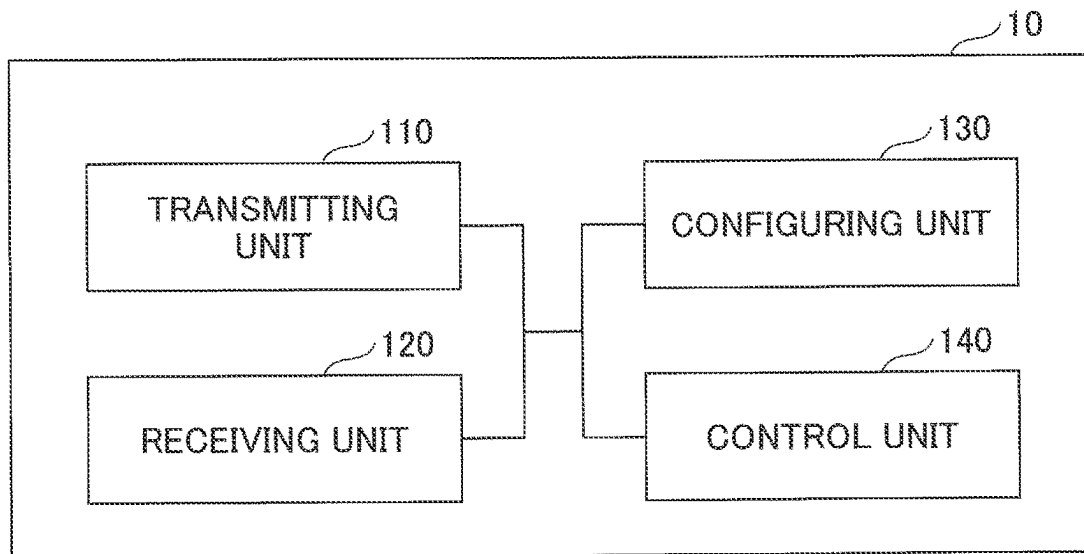


FIG.22

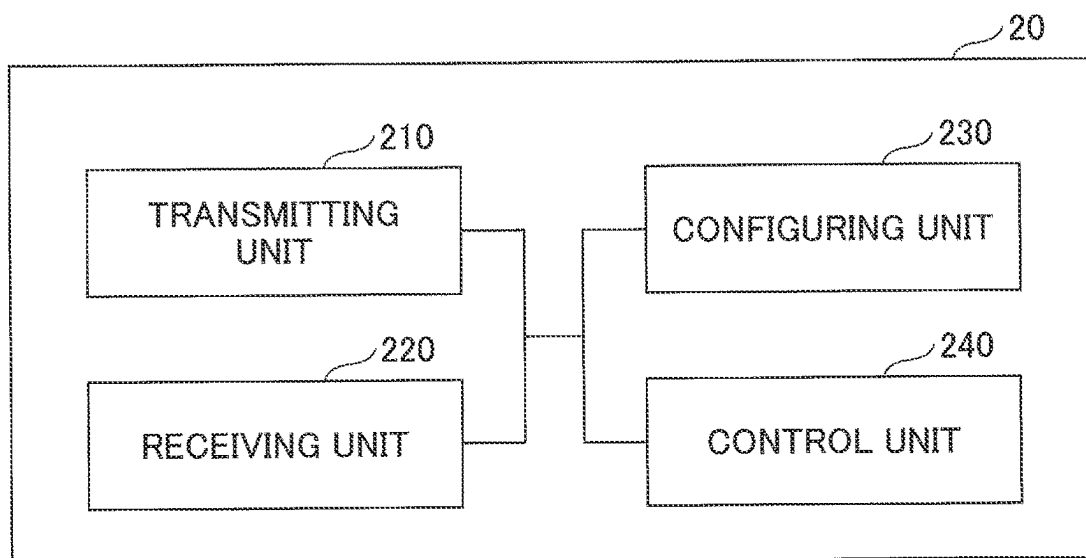
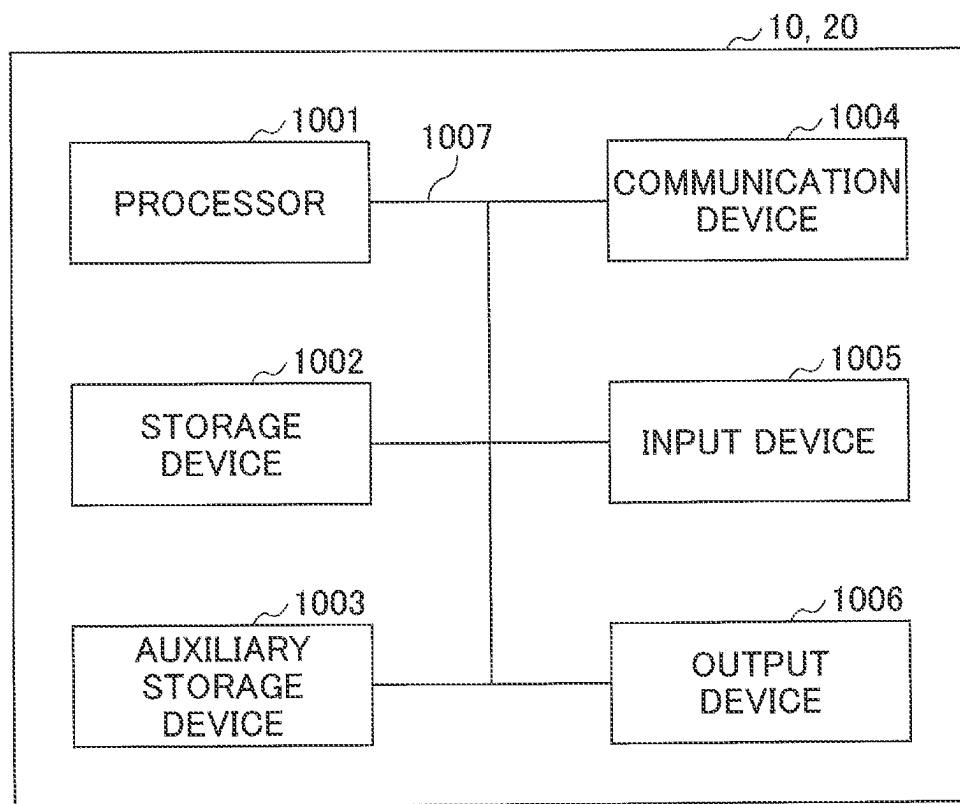


FIG.23



TERMINAL AND COMMUNICATION METHOD

TECHNICAL FIELD

The present invention relates to a terminal and a communication method in a radio communication system.

BACKGROUND ART

In LTE (Long Term Evolution) and a successor system of the LTE (for example, LTE-A (LTE-Advanced) and NR (New Radio) (which is also referred to as 5G)), D2D (Device to Device) techniques where terminals perform direct communication with each other without involving a base station are discussed (for example, non-patent document 1).

The D2D reduces traffic between terminals and base stations, and even if the base stations are unable to perform communications in the event of disasters, the D2D enables communication between the terminals. Note that although the D2D is referred to as “sidelink” in 3GPP (3rd Generation Partnership Project), the D2D is used as a more generic terminology in the present specification. However, the sidelink may be used in descriptions of embodiments as stated below if necessary.

The D2D communication is roughly divided into: D2D discovery for discovering other terminals capable of communication; and D2D communication (also referred to as direct communication between terminals or the like) for direct communication between terminals. In the following, when the D2D communication, the D2D discovery or the like are not particularly distinguished, they are simply referred to as D2D. Also, signals transmitted and received in the D2D are referred to as D2D signals. Various use cases of services associated with V2X (Vehicle to Everything) in the NR are being discussed (for example, non-patent document 2).

CITATION LIST

Non-Patent Literature

- [NPTL 1] 3GPP TS 38.211 V16.0.0 (2019-12)
[NPTL 2] 3GPP TR 22.886 V15.1.0 (2017-03)

SUMMARY OF INVENTION

Technical Problem

Direct terminal-to-terminal communication in NR-V2X supports the resource allocation mode 2. When the resource allocation mode 2 is applied, the terminal performs sensing and selects the available (usable) resource candidates from a resource selection window based on the sensing result.

Here, it is assumed that partial sensing (partial sensing) is used to reduce power consumption when resource allocation mode 2 is applied. On the other hand, NR sidelink supports aperiodic traffic in addition to periodic traffic, and, therefore, if partial sensing is used to reduce the sensing targets, it is necessary to change the operation in conventional partial sensing.

The present invention has been made in view of the above points, and an object of the present invention is to reduce the sensing targets when the resources used by the terminal are autonomously selected, in direct terminal-to-terminal communication.

Solution to Problem

According to the disclosed technology, there is provided a terminal including a receiving unit configured to perform sensing of resource reservation information and resource allocation information transmitted from another terminal by using a sensing window; a control unit configured to determine an available resource set, by identifying a predetermined resource in a resource selection window, based on the resource reservation information and the resource allocation information; and a transmitting unit configured to execute transmission to the another terminal by using a resource selected from the resource set.

Advantageous Effects of Invention

According to the disclosed technology, in direct terminal-to-terminal communication, when the resources used by the terminal are autonomously selected, the sensing targets can be reduced.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is a diagram illustrating V2X;

FIG. 2 is a diagram illustrating an example (1) of a transmission mode of the V2X;

FIG. 3 is a diagram illustrating an example (2) of a transmission mode of the V2X;

FIG. 4 is a diagram illustrating an example (3) of a transmission mode of the V2X;

FIG. 5 is a diagram illustrating an example (4) of a transmission mode of the V2X;

FIG. 6 is a diagram illustrating an example (5) of a transmission mode of the V2X;

FIG. 7 is a diagram illustrating an example (1) of a communication type of the V2X;

FIG. 8 is a diagram illustrating an example (2) of a communication type of the V2X;

FIG. 9 is a diagram illustrating an example (3) of a communication type of the V2X;

FIG. 10 is a sequence diagram illustrating an operation example (1) of the V2X;

FIG. 11 is a sequence diagram illustrating an operation example (2) of the V2X;

FIG. 12 is a sequence diagram illustrating an operation example (3) of the V2X;

FIG. 13 is a sequence diagram illustrating an operation example (4) of the V2X;

FIG. 14 is a diagram illustrating an example (1) of a transmission operation;

FIG. 15 is a diagram illustrating an example (2) of a transmission operation;

FIG. 16 is a diagram illustrating an example (3) of a transmission operation;

FIG. 17 is a flowchart illustrating an example (1) of sensing according to an embodiment of the present invention;

FIG. 18 is a diagram illustrating an example (1) of sensing according to an embodiment of the present invention;

FIG. 19 is flowchart illustrating an example (2) of sensing according to an embodiment of the present invention;

FIG. 20 is a diagram illustrating an example (2) of sensing according to an embodiment of the present invention;

FIG. 21 is a diagram illustrating an example of a functional configuration of a base station 10 according to an embodiment of the present invention;

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FIG. 22 is a diagram illustrating an example of a functional configuration of a terminal 20 according to an embodiment of the present invention; and

FIG. 23 is a diagram illustrating an example of a hardware configuration of the base station 10 or the terminal 20 according to an embodiment of the present invention.

DESCRIPTION OF EMBODIMENT

Hereinafter, embodiments of the present invention will be described with reference to the drawings. The embodiments described below are merely examples, and embodiments to which the present invention is applied are not limited to the following embodiments.

In operations of a radio communication system of an embodiment of the present invention, conventional techniques are used as needed. Note that the conventional techniques are conventional LTE, for example, but are not limited to the conventional LTE. Also, unless specifically stated otherwise, it should be appreciated that the terminology "LTE" used herein has a broader meaning including LTE-Advanced, its subsequent schemes (e.g., NR) or a wireless LAN (Local Area Network).

In addition, in the embodiment of the invention, a duplexing scheme may be Time Division Duplexing (TDD), may be Frequency Division Duplexing (FDD), or may be other schemes (for example, Flexible Duplexing or the like).

Also, in embodiments of the present invention, "configuring" a radio parameter or the like may mean that a predetermined value is preconfigured or that a radio parameter indicated by the base station 10 or the terminal 20 is configured.

FIG. 1 is a diagram illustrating V2X. In 3GPP, it is considered that V2X (Vehicle to Everything) or eV2X (enhanced V2X) are implemented through enhancement of D2D functions, and the specifications are being developed. As illustrated in FIG. 1, the V2X is a part of ITS (Intelligent Transport Systems) and is a collective term for V2V (Vehicle to Vehicle), which means the form of communication between vehicles, V2I (Vehicle to Infrastructure), which means the form of communication between a vehicle and a RSU (Road-Side Unit) located at a roadside, V2N (Vehicle to Network), which means the form of communication between a vehicle and an ITS server, and V2P (Vehicle to Pedestrian), which means the form of communication between a vehicle and a mobile terminal carried by a pedestrian.

Also, V2X utilizing cellular communication and inter-terminal communication of LTE or NR is being discussed in the 3GPP. The V2X utilizing cellular communication may be referred to as cellular V2X. In the V2X of the NR, implementations of large capacity, low delay, high reliability and QoS (Quality of Service) control are discussed.

It is expected that discussions of V2X of LTE or NR is not limited to the 3GPP specifications in the future. For example, it is expected that interoperability, cost reduction by implementation of an upper layer, combination or switching manner of multiple RATs (Radio Access Technology), regulatory compliance in respective countries, data acquisition, distribution, database management and use of V2X platforms of LTE or NR will be discussed.

In embodiments of the present invention, a communication device is mainly assumed to be installed in a vehicle, the embodiments of the present invention are not limited to those embodiments. For example, the communication device may be a terminal carried by a person, a device

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installed in a drone or an aircraft, a base station, an RSU, a relay node, a terminal having scheduling capabilities and so on.

Note that a SL (Sidelink) may be differentiated from an UL (Uplink) or a DL (Downlink), based on one of, or combinations of, 1) to 4) below. Also, the SL may be referred to as other names.

- 1) Resource arrangement of a time domain
- 2) Resource arrangement of a frequency domain
- 3) Synchronization signal to be referenced (including SLSS (Sidelink Synchronization Signal))
- 4) Reference signal used for pathloss measurement for transmit power control

Also, regarding OFDM (Orthogonal Frequency Division Multiplexing) of the SL or the UL, any of CP-OFDM (Cyclic-Prefix OFDM), DFT-S-OFDM (Discrete Fourier Transform-Spread-OFDM), OFDM without transform precoding or OFDM with transform precoding may be applied.

In the SL of the LTE, Mode 3 and Mode 4 regarding SL resource allocation to a terminal 20 are defined. In Mode 3, transmission resources are dynamically allocated by DCI (Downlink Control Information) transmitted from a base station 10 to the terminal 20. Also, SPS (Semi Persistent Scheduling) is enabled in Mode 3. In Mode 4, the terminal 20 autonomously selects transmission resources from a resource pool.

Note that a slot according to an embodiment of the present invention may be replaced with a symbol, a mini-slot, a subframe, a radio frame or a TTI (Transmission Time Interval). Also, a cell according to an embodiment of the present invention may be replaced with a cell group, a carrier component, a BWP, a resource pool, a resource, a RAT (Radio Access Technology), a system (including a wireless LAN) or the like.

Note that in the embodiment of the present invention, the terminal 20 is not limited to the V2X terminal, but may be any type of terminal that performs D2D communication. For example, the terminal 20 may be a terminal carried by a user, such as a smartphone, or an IoT (Internet of Things) device, such as a smart meter.

FIG. 2 is a diagram illustrating an example (1) of a transmission mode for the V2X. In the transmission mode for sidelink communication illustrated in FIG. 2, at step 1, the base station 10 transmits a scheduling for a sidelink to the terminal 20A. Then, the terminal 20A transmits a PSCCH (Physical Sidelink Control Channel) and a PSSCH (Physical Sidelink Shared Channel) to the terminal 20B based on the received scheduling (step 2). The transmission mode of the sidelink communication illustrated in FIG. 2 may be referred to as sidelink transmission mode 3 for the LTE. In the sidelink transmission mode 3 for the LTE, Uu based sidelink scheduling is performed. The Uu means a radio interface between a UTRAN (Universal Terrestrial Radio Access Network) and a UE (User Equipment). Note that the transmission mode for the sidelink communication illustrated in FIG. 2 may be referred to as sidelink transmission mode 1 for the NR.

FIG. 3 is a diagram illustrating an example (2) of a transmission mode for the V2X. In the transmission mode for the sidelink communication illustrated in FIG. 3, at step 1, the terminal 20A uses autonomously selected resource to transmit a PSCCH and a PSSCH to the terminal 20B.

The transmission mode for the sidelink communication illustrated in FIG. 3 may be referred to as sidelink transmission mode 4 for the LTE. In sidelink transmission mode 4 for the LTE, the UE itself perform resource selection.

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FIG. 4 is a diagram illustrating an example (3) of a transmission mode for the V2X. In a transmission mode for the sidelink communication illustrated in FIG. 4, at step 1, the terminal 20A uses an autonomously selected resource to transmit a PSCCH and a PSSCH to the terminal 20B. Analogously, the terminal 20B uses an autonomously selected resource to transmit a PSCCH and a PSSCH to the terminal 20A (step 1). The transmission mode for the sidelink communication illustrated in FIG. 4 may be referred to as sidelink transmission mode 2a for the NR. In the sidelink transmission mode 2 for the NR, the terminal 20 itself performs resource selection.

FIG. 5 is a diagram illustrating an example (4) of a transmission mode for the V2X. In the transmission mode for the sidelink communication illustrated in FIG. 5, at step 0, the base station 10 transmits a sidelink grant to the terminal 20A via an RRC (Radio Resource Control) configuration. Then, the terminal 20A transmits a PSSCH to the terminal 20B via received resource pattern (step 1). The transmission mode for the sidelink communication illustrated in FIG. 5 may be referred to as sidelink transmission mode 2c for the NR.

FIG. 6 is a diagram illustrating an example (5) of a transmission mode for the V2X. In a transmission mode for the sidelink communication illustrated in FIG. 6, at step 1, the terminal 20A transmits a sidelink scheduling to the terminal 20B via a PSCCH. Then, the terminal 20B transmits a PSSCH to the terminal 20A based on the received scheduling (step 2). The transmission mode for the sidelink communication illustrated in FIG. 6 may be referred to as sidelink transmission mode 2d for the NR.

FIG. 7 is a diagram illustrating an example (1) of a communication type of the V2X. The sidelink communication type illustrated in FIG. 7 is a unicast. The terminal 20A transmits a PSCCH and a PSSCH to the terminal 20. In the example illustrated in FIG. 7, the terminal 20A performs unicast for the terminal 20B and also performs a unicast for the terminal 20C.

FIG. 8 is a diagram illustrating an example (2) of a communication type for the V2X. The sidelink communication type illustrated in FIG. 8 is a group cast. The terminal 20A transmits a PSCCH and a PSSCH to a group to which one or more terminals 20 belong. In the example illustrated in FIG. 8, the group includes the terminals 20B and 20C, and the terminal 20A performs a group cast to the group.

FIG. 9 is a diagram illustrating an example (3) of a communication type for the V2X. The sidelink communication type illustrated in FIG. 9 is a broadcast. The terminal 20A transmits a PSCCH and a PSSCH to one or more terminals 20. In the example illustrated in FIG. 9, the terminal 20A performs a broadcast to the terminals 20B, 20C and 20D. Note that the terminal 20A illustrated in FIGS. 7 to 9 may be referred to as a header UE.

Also, it is assumed in NR-V2X that a HARQ (Hybrid Automatic Repeat Request) is supported for the sidelink unicast and group cast. In addition, SFCI (Sidelink Feedback Control Information) including a HARQ response is defined in the NR-V2X. In addition, it is being discussed that the SFCI is transmitted via a PSFCH (Physical Sidelink Feedback Channel).

Although the PSFCH is used for sidelink transmission of a HARQ-ACK in descriptions below, it is merely an example. For example, a PSCCH may be used to transmit the sidelink HARQ-ACK, a PSSCH may be used to transmit the sidelink HARQ-ACK or other channels may be used to transmit the sidelink HARQ-ACK.

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In the following, information reported by the terminal 20 in HARQs is referred to as the HARQ-ACK in general for convenience. The HARQ-ACK may be referred to as HARQ-ACK information. Also, more specifically, a codebook applied to the HARQ-ACK information reported from the terminal 20 to the base station 10 or the like is referred to as a HARQ-ACK codebook. The HARQ-ACK codebook specifies bit sequences of the HARQ-ACK information. Note that not only ACK but also NACK is transmitted in the HARQ-ACK.

FIG. 10 is a diagram illustrating an example (1) of arrangement and operation of a radio communication system according to an embodiment of the present invention. As illustrated in FIG. 10, the radio communication system according to an embodiment of the present invention has terminals 20A and 20B. Note that although large numbers of user equipment are present in the real world, terminals 20A and 20B are exemplarily illustrated in FIG. 10.

In the following, if the terminals 20A, 20B and the like are not particularly distinguished, they are simply described as “terminal 20” or “user equipment”. Although the case where both the terminals 20A and 20B are within a coverage of a cell is illustrated in FIG. 10, the operation according to an embodiment of the present invention may be also applied to the case where the terminal 20B is out of the coverage.

As stated above, in the present embodiment, the terminal 20 is a device installed in a vehicle such as a car, for example, and has cellular communication functions and sidelink functions as a UE in the LTE or the NR. The terminal 20 may be a generic mobile terminal (such as a smartphone). Also, the terminal 20 may be a RSU. The RSU may be a UE type of RSU having UE functions or gNB type of RSU having functions of a base station apparatus.

Note that the terminal 20 is not necessarily a device of a single housing, and in a case where various sensors are distributed and installed in a vehicle, for example, the terminal 20 may be a device including those sensors.

Also, processing contents at the terminal 20 for sidelink transmission data are basically similar to those for UL transmission in the LTE or the NR. For example, the terminal 20 scrambles and modulates a codeword of transmission data to generate complex-valued symbols and maps the complex-valued symbols (transmission signal) to one or two layers for precoding. Then, the terminal 20 maps the precoded complex-valued symbols to a resource element to generate a transmission signal (for example, a complex-valued time-domain SC-FDMA signal), and performs transmission from each antenna port.

Note that the base station 10 has cellular communication functions to serve as a base station for the LTE or the NR, and functions (for example, resource pool configuration, resource allocation or the like) that enable communication with the terminal 20 according to the present embodiment. Also, the base station 10 may be an RSU (a gNB type of RSU).

Also, a signal waveform utilized by the terminal 20 in the SL or the UL in the radio communication system according to an embodiment of the present invention, may be OFDMA, SC-FDMA or others.

At step S101, the terminal 20A autonomously selects a resource used for a PSCCH and a PSSCH from a resource selection window having a predetermined period. The resource selection window may be configured to the terminal 20 by the base station 10.

At steps S102 and S103, the terminal 20A uses the resource autonomously selected at step S101 to transmit SCI (Sidelink Control Information) in a PSCCH and SL data in

a PSSCH. For example, the terminal **20A** may use a time resource that is the same as a time resource of the PSSCH, and a frequency resource that is adjacent to a frequency resource of the PSSCH, to transmit the SCI (PSCCH).

The terminal **20B** receives the SCI (PSCCH) and the SL data (PSSCH) transmitted from the terminal **20A**. The SCI received in the PSCCH may include PSFCH resource information used for transmitting a HARQ-ACK in response to reception of the data. The terminal **20A** may include information of an autonomously selected resource in the SCI, and transmit the SCI.

At step **S104**, the terminal **20B** uses the PSFCH resource indicated by the received SCI, to transmit the HARQ-ACK for the received data, to the terminal **20A**.

At step **S105**, if the HARQ-ACK received at step **S104** indicates a request for retransmission, that is, if it is a NACK (negative response), the terminal **20A** retransmits the PSCCH and PSSCH to the terminal **20B**. The terminal **20A** may use the autonomously selected resource to retransmit the PSCCH and PSSCH.

Note that if the HARQ control is not performed, steps **S104** and **S105** need not be performed.

FIG. **11** is a diagram illustrating an example (2) of arrangement and operation of the radio communication system according to an embodiment of the present invention. Blind retransmission that does not rely on the HARQ control, may be performed so as to improve a transmission success rate or a reachable distance.

At step **S201**, the terminal **20A** autonomously selects a resource to be used for a PSCCH and a PSSCH, from a resource selection window having a predetermined period. The resource selection window may be configured to the terminal **20** by the base station **10**.

At steps **S202** and **S203**, the terminal **20A** uses the resource autonomously selected at step **S201** to transmit SCI via the PSCCH and SL data via the PSSCH. For example, the terminal **20A** may use a time resource that is the same as a time resource of the PSSCH, and a frequency resource that is adjacent to a frequency resource of the PSSCH, to transmit the SCI (PSCCH).

At step **S204**, the terminal **20A** uses the resource autonomously selected at step **S201** to retransmit the SCI via the PSCCH and the SL data via the PSSCH to the terminal **20B**. The retransmission at step **S204** may be performed multiple times.

Note that if the blind retransmission is not performed, step **S204** need not be performed.

FIG. **12** is a diagram illustrating an example (3) of arrangement and operation of the radio communication system according to an embodiment of the present invention. The base station **10** may perform sidelink scheduling. That is, the base station **10** may determine a resource to be used for sidelink by the terminal **20**, and transmit information indicative of the resource to the terminal **20**. In addition, if the HARQ control is applied, the base station **10** may transmit information indicative of a PSFCH resource to the terminal **20**.

At step **S301**, the base station **10** performs an SL scheduling for the terminal **20A** by transmitting DCI (Downlink Control Information) in a PDCCH. Hereinafter, for convenience, the DCI for the SL scheduling is referred to as SL scheduling DCI.

Also, at step **S301**, it is assumed that the base station **10** also transmits the DCI for a DL scheduling (may be referred to as DL allocation) in the PDCCH. Hereinafter, for convenience, the DCI for the DL scheduling is referred to as DL scheduling DCI. Upon receiving the DL scheduling DCI, the

terminal **20A** uses a resource indicated by the DL scheduling DCI to receive DL data in a PDSCH.

At steps **S302** and **S303**, the terminal **20A** uses a resource indicated by the SL scheduling DCI to transmit SCI (Sidelink Control Information) in a PSCCH and SL data in a PSSCH. Note that only a resource of the PSSCH may be indicated by the SL scheduling DCI. In this case, for example, the terminal **20A** may use a time resource that is same as a time resource of the PSSCH, and a frequency resource that is adjacent to a frequency resource of the PSSCH, to transmit the SCI (PSCCH).

The terminal **20B** receives the SCI (PSCCH) and SL data (PSSCH) transmitted from the terminal **20A**. The SCI received in the PSCCH includes information regarding a resource of a PSFCH for transmitting a HARQ-ACK in response to reception of the data from the terminal **20B**.

The information regarding the resource is included in the DL scheduling DCI or the SL scheduling DCI transmitted from the base station **10** at step **S301**, and the terminal **20A** acquires the information regarding the resource from the DL scheduling DCI or the SL scheduling DCI, and the terminal **20A** includes the acquired information in the SCI. Alternatively, the information regarding the resource need not be included in the DCI transmitted from the base station **10**, and the terminal **20A** may autonomously include the information regarding the resource in the SCI, and transmit the SCI.

At step **S304**, the terminal **20B** uses a resource of the PSFCH indicated by the received SCI to transmit a HARQ-ACK in response to the received data to the terminal **20A**.

At step **S305**, the terminal **20A** transmits the HARQ-ACK at a timing (for example, a timing in unit of slot) indicated by the DL scheduling DCI (or the SL scheduling DCI), by using a PUCCH (Physical Uplink Control Channel) resource indicated by the DL scheduling DCI (or the SL scheduling DCI), and the base station **10** receives the HARQ-ACK. A codebook of the HARQ-ACK may include: the HARQ-ACK received from the terminal **20B**; and the HARQ-ACK in response to DL data. Note that the HARQ-ACK in response to the DL data need not be included in the case where, for example, no DL data is allocated.

Note that if HARQ control is not performed, steps **S304** and **S305** need not be performed.

FIG. **13** is a diagram illustrating an operation example (4) according to an embodiment of the present invention. As stated above, transmission of the HARQ response via the PSFCH is supported in the NR sidelink. Note that a format similar to PUCCH (Physical Uplink Control Channel) format 0 is available as the PSFCH format. In other words, in the PSFCH format, the PRB (Physical Resource Block) size is equal to 1, and the ACK and the NACK may be a sequence based format that can be differentiated based on the sequence difference. The PSFCH format is not limited to the above. A resource of the PSFCH may be placed to at the last symbol of a slot, or at multiple symbols at the end of a slot. Also, the cycle N may be configured for the PSFCH resource, or may be specified in advance. The cycle N may be configured or specified in advance, in unit of slot.

In FIG. **13**, the vertical axis corresponds to a frequency domain, and the horizontal axis corresponds to a time domain. The PSCCH may be placed at first symbol of a slot, at multiple symbols from the first symbol of a slot, or at multiple symbols from a symbol other than the first symbol of a slot. The PSFCH may be placed at the last symbol of a slot, or at multiple symbols at the end of a slot. In the example illustrated in FIG. **13**, three sub-channels are configured in a resource pool, and the two PSFCHs are placed at the third slot from a slot where the PSSCH is placed. The

arrows from the PSSCH to the PSFCHs shows examples of the PSFCHs associated with the PSSCH.

If the HARQ response in NR-V2X group cast is option 2 of transmitting the ACK or NACK, a resource used for transmission and reception of the PSFCH, must be determined. As illustrated in FIG. 13, at step S401, the terminal 20A serving as the transmitting terminal 20 performs group cast to the terminals 20B, 20C and 20D serving as the receiving terminals 20, via an SL-SCH. Then, at step S402, the terminal 20B, 20C and 20D use a PSFCH #B, a PSFCH #C, and a PSFCH #D, respectively, to transmit the HARQ responses to the terminal 20A. Here, as illustrated in the example in FIG. 13, if the number of available PSFCH resources is less than the number of receiving terminals 20 belonging to a group, how to allocate the PSFCH resources, must be determined. Note that the transmitting terminal 20 may acquire the number of receiving terminals 20 in the group cast.

FIG. 14 is a diagram illustrating an example (1) of a transmission operation. In the resource allocation mode 2, the terminal 20 selects a resource and performs transmission. As illustrated in FIG. 14, the terminal 20 performs sensing in a sensing window in the resource pool. By sensing, the terminal 20 receives a resource reservation field or resource assignment field included in the SCI transmitted from another terminal 20 and identifies an available (usable) resource candidate in the resource selection window in the resource pool, based on the fields. Subsequently, the terminal 20 randomly selects resources from the available resource candidates.

Further, the configuration of the resource pool may have a period, as illustrated in FIG. 14. For example, the period is referred to as a hyperframe and may be a time period of 10240 milliseconds. FIG. 14 is an example in which slot t_0^{SL} to slot $t_{T_{max}}^{SL}$ are configured as a resource pool. For the resource pool in the hyperframe, areas may be configured by a bitmap, for example.

Further, as illustrated in FIG. 14, the transmission trigger at the terminal 20 occurs at the slot n , and it is assumed that the priority of the transmission is p_{TX} . The terminal 20 can detect, for example, that another terminal 20 is performing transmission with a priority p_{RX} in the sensing window from the slot $n-T_0$ to a slot immediately before the slot $n-T_{proc,0}$. If the SCI is detected in the sensing window and the RSRP (Reference Signal Received Power) exceeds the threshold, a resource in the resource selection window corresponding to the SCI is excluded. Further, if the SCI is detected in the sensing window and the RSRP is below the threshold, a resource in the resource selection window corresponding to the SCI is not excluded. The threshold may be, for example, a threshold $Th_{p_{TX},p_{RX}}$, which is configured or defined for each resource in the sensing window, based on priority p_{TX} and priority p_{RX} .

Further, as in the slot t_m^{SL} illustrated in FIG. 14, a resource in the resource selection window that is to be a candidate for resource reservation information corresponding to a resource in the sensing window that has not been monitored, for example, due to transmission, is excluded.

In the resource selection window from slot $n+T_1$ to slot $n+T_2$, the resources occupied by another UE are identified as illustrated in FIG. 14, and the resources, from which the identified resources have been excluded, become the available resource candidates. When the set of available resource candidates is S_A , if S_A is less than 20% of the resource selection window, then the threshold $Th_{p_{TX},p_{RX}}$ configured for each resource in the sensing window may be raised by 3 dB, and the identification of resources may be performed

again. That is, by raising the threshold $Th_{p_{TX},p_{RX}}$ and performing resource identification again, the resources, which are not excluded because RSRP is below the threshold, are increased, and the resource candidate set S_A may become 20% or more of the resource selection window. If S_A is less than 20% of the resource selection window, the operation of raising the threshold $Th_{p_{TX},p_{RX}}$ configured for each resource in the sensing window by 3 dB to perform resource identification again, may be repeated.

The lower layer of the terminal 20 may report the S_A to the higher layer. The higher layer of the terminal 20 may perform random selection with respect to the S_A to determine a resource to be used. The terminal 20 may perform sidelink transmission by using the determined resource.

Here, power saving by random resource selection and partial sensing is being discussed in the NR release 17 sidelink. For example, for power saving, random resource selection and partial sensing of the sidelink in the LTE release 14 may be applied to the resource allocation mode 2 of the NR release 16 sidelink.

FIG. 15 is a diagram illustrating an example (2) of a transmission operation. In the LTE sidelink, if partial sensing is not configured by the higher layer, the terminal 20 selects a resource and performs transmission, as illustrated in FIG. 15. As illustrated in FIG. 15, the terminal 20 performs sensing in a sensing window in the resource pool. By performing the sensing, the terminal 20 receives a resource reservation field included in the SCI transmitted from another terminal 20 and identifies the available resource candidates within the resource selection window in the resource pool based on the field. Subsequently, the terminal 20 randomly selects a resource from the available resource candidates.

Further, the configuration of the resource pool may include a period, as illustrated in FIG. 15. For example, the period is referred to as a hyperframe and may be a time period of 10240 milliseconds. FIG. 15 is an example in which the subframe t_0^{SL} to the subframe $t_{T_{max}}^{SL}$ are configured as a resource pool. For the resource pool in the hyperframe, areas may be configured, for example, by a bitmap.

Further, as illustrated in FIG. 15, the transmission trigger at the terminal 20 occurs at the subframe n , and the priority of the transmission is p_{TX} . The terminal 20 can detect, for example, that another terminal 20 is performing transmission with the priority p_{RX} in the sensing window from the subframe $t_{n-10 \times P_{step}}^{SL}$ to the subframe t_{n-1}^{SL} . If the SCI is detected in the sensing window and the RSRP is above the threshold, the resource in the resource selection window corresponding to the SCI is excluded. Further, if the SCI is detected in the sensing window and the RSRP is below the threshold, the resource in the resource selection window corresponding to the SCI is not excluded. The threshold may be, for example, a threshold $Th_{p_{TX},p_{RX}}$, which is configured or defined for each resource in the sensing window based on priority p_{TX} and priority p_{RX} .

Further, as in the subframe t_z^{SL} illustrated in FIG. 15, a resource in the resource selection window that is to be candidate for resource reservation information corresponding to a resource in the sensing window that has not been monitored, for example, due to transmission, is excluded.

In the resource selection window from subframe $n+T_1$ to subframe $n+T_2$, the resources occupied by another UE are identified as illustrated in FIG. 15, and the resources, from which the identified resources have been excluded, become available resource candidates. When the set of available resource candidates is S_A , and if S_A is less than 20% of the

resources in the resource selection window, the threshold $Th_{pTX,pRX}$ configured for each resource in the sensing window may be raised by 3 dB, and the identification of resources may be performed again. That is, by raising the threshold $Th_{pTX,pRX}$ and performing resource identification again, the resources, which are not excluded because RSRP is below the threshold, may be increased. Further, the RSSI of each resource in the S_A may be measured and the resource with the minimum RSSI may be added to the set S_B . The operation of adding the resource with the minimum RSSI included in the set S_A to the set S_B may be repeated until the set S_B of resource candidates becomes greater than or equal to 20% of the resource selection window.

The lower layer of the terminal **20** may report the S_B to the higher layer. The higher layer of the terminal **20** may perform random selection with respect to the S_3 to determine the resource to be used. The terminal **20** may perform sidelink transmission using the determined resource. Note that the terminal **20** may obtain a resource and then may use the resource periodically without performing sensing for a predetermined number of times (for example, C_{resel} times).

FIG. 16 is a diagram illustrating an example (3) of a transmission operation. In the LTE sidelink, when the partial sensing is configured by the higher layer, the terminal **20** selects a resource and performs transmission, as illustrated in FIG. 16. As illustrated in FIG. 16, the terminal **20** performs partial sensing in a sensing window in the resource pool. By performing partial sensing, the terminal **20** receives the resource reservation field included in the SCI transmitted from another terminal **20** and identifies available resource candidates within the resource selection window in the resource pool, based on the field. Subsequently, the terminal **20** randomly selects a resource from the available resource candidates.

Further, the configuration of the resource pool may include a period, as illustrated in FIG. 16. For example, the period is referred to as a hyperframe and may be a time period of 10240 milliseconds. FIG. 16 is an example in which the subframe t_0^{SL} to the subframe t_{Tmax}^{SL} are configured as a resource pool. For the resource pool in the hyperframe, areas may be configured, for example, by a bitmap.

As illustrated in FIG. 16, the transmission trigger at the terminal **20** occurs at the subframe n , and the priority of the transmission is assumed to be p_{TX} . As illustrated in FIG. 16, Y subframes from subframe t_y^{SL} to subframe t_{y+Y}^{SL} , among the subframes from subframe $n+T_1$ to subframe $n+T_2$, may be configured as a resource selection window. The Y subframes may be non-contiguous subframes.

The terminal **20** may detect, for example, that another terminal **20** is performing the transmission with the priority p_{RX} in one or more sensing windows from the subframe $t_{y-k \times Pstep}^{SL}$ to the subframe $t_{y+k \times Pstep}^{SL}$ having a Y subframe length. The aforementioned k may be, for example, a bitmap of 10 bits. FIG. 16 illustrates an example in which the third and sixth bits of the bitmap k are configured to be "1" indicating that partial sensing is to be performed. That is, in FIG. 16, the subframe $t_{y-6 \times Pstep}^{SL}$ to the subframe $t_{y+Y-6 \times Pstep}^{SL}$, and the subframe $t_{y-3 \times Pstep}^{SL}$ to the subframe $t_{y+Y-3 \times Pstep}^{SL}$ are configured as sensing windows. As noted above, the i-th bit of bitmap k may correspond to a sensing window from subframe $t_{y-1 \times Pstep}^{SL}$ to subframe $t_{y+Y-1 \times Pstep}^{SL}$.

If the SCI is detected in one or more of the above sensing windows and the RSRP is above the threshold, resources in the resource selection window corresponding to the SCI are excluded. Further, if the SCI is detected in the sensing

window and the RSRP is below the threshold, resources in the resource selection window corresponding to the SCI are not excluded. The threshold may be, for example, a threshold $Th_{pTX,pRX}$, which is configured or defined for each resource in the sensing window based on priority p_{TX} and priority p_{RX} .

In the resource selection window in which the Y subframes are configured, the terminal **20** identifies a resource occupied by another UE, and the resources, from which the identified resource has been excluded, become the available resource candidates. When the set of available resource candidates is S_A , if S_A is less than 20% of the resources in the resource selection window, the threshold $Th_{pTX,pRX}$ configured for each resource in the sensing window may be raised by 3 dB to perform resource identification again. That is, by raising the threshold $Th_{pTX,pRX}$ and performing resource identification again, the resources that are not excluded because RSRP is below the threshold, may be increased. Further, the RSSI of each resource in the S_A may be measured and the resource with the minimum RSSI may be added to the set S_B . The operation of adding the resource with the minimum RSSI included in the set S_A to the set S_B may be repeated until the set S_B of resource candidates becomes greater than or equal to 20% of the resource selection window.

The lower layer of the terminal **20** may report the S_B to the higher layer. The higher layer of the terminal **20** may perform random selection with respect to the S_B to determine a resource to be used. The terminal **20** may perform sidelink transmission using the determined resource. Note that the terminal **20** may obtain a resource and then use the resource periodically without performing sensing for a predetermined number of times (for example, C_{resel} times).

The partial sensing in the LTE sidelink illustrated in FIG. 16 realizes energy saving by reducing the sensing targets, based on the assumption of periodic traffic. On the other hand, aperiodic traffic is supported in the NR sidelink, and, therefore, when partial sensing in the LTE sidelink is applied to the NR sidelink, sensing of aperiodic traffic becomes difficult, and the performance may be degraded.

Thus, a method of resource selection that does not degrade the performance while saving power by executing sensing to accommodate NR sidelink traffic and reducing the sensing targets, will be described below. For example, the resource candidates and/or resource selection candidates to be targets of sensing, may be narrowed down. Further, instructions may be configured (in advance) to prohibit specific operations, such as, for example, preventing PSCCH decoding. Note that, in the embodiment of the present invention, a resource allocation mode, in which the terminal selects the resources by itself, is assumed, and, for example, a resource allocation mode 2 may be assumed.

FIG. 17 is a flowchart illustrating an example (1) of sensing according to an embodiment of the present invention. FIG. 18 is a diagram illustrating an example (1) of sensing according to an embodiment of the present invention. As illustrated in FIGS. 17 and 18, the terminal **20** may apply two types of sensing, each of which may have a different window, and, for example, the two types of sensing, the sensing using the resource reservation field and the sensing using the resource allocation field, may be performed. Note that as illustrated in FIG. 18, it is assumed that the transmission trigger at the terminal **20** occurs at the subframe n , and the priority of the transmission is p_{TX} . In the sensing example (1) illustrated in FIGS. 17 and 18, energy

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saving can be achieved by reducing the number of resource candidates and/or resource selection candidates to be the target of sensing.

In the following description, when slot n_2 is located later than slot n_1 in the time domain, the section [slot n_1 , slot n_2] represents the section from slot n_1 to slot n_2 including slot n_1 and slot n_2 , and the section [slot n_1 , slot n_2) represents the section from slot n_1 to a slot immediately before slot n_2 including slot n_1 and not including slot n_2 .

In step S11, the terminal 20 may assume Y slots from t_y^{SL} to t_{y+Y}^{SL} within a section [slot $n+T_1$, $n+T_2$] as a resource selection window, as illustrated in FIG. 18. The Y slots may be non-contiguous slots.

In step S12, the terminal 20 may assume that, all or a part of the slots by which the Y slots can be reserved, for example, as periodic traffic according to the resource reservation field, are the sensing window #1. That is, the terminal 20 may perform partial sensing of periodic traffic. The determination of the sensing window #1 is made based on the resource reservation field, but the sensing operation, which uses the SCI received in the sensing window #1, may be performed based on the resource reservation field or may be performed based on the resource allocation field. For example, the terminal 20 may assume that a section $[t_{y-k \times A}^{SL}, t_{y+Y-k \times A}^{SL}]$ is the sensing window #1. The aforementioned k may be a bitmap that is X bits long. The aforementioned A may be a value that may take up to a maximum of 16 different values that can be specified by the resource reservation field. The sensing window #1 may be included in the section $[T_0, n-T_{proc,0}]$.

In the example illustrated in FIG. 18, the third and sixth bits of the bitmap k are configured to "1" indicating that partial sensing is to be performed, and the sensing window #1 is configured in a section $[t_{y-6 \times A}^{SL}, t_{y+Y-6 \times A}^{SL}]$ and in a section $[t_{y-3 \times A}^{SL}, t_{y+Y-3 \times A}^{SL}]$.

The parameters Y, k, A, and X may be configured or predefined and may be the parameters indicated to the terminal 20 from the base station 10.

In step S13, the terminal 20 may assume that, all or a part of the slots in which the Y slots can be reserved, for example, as aperiodic traffic according to the resource allocation field, are the sensing window #2. For example, the terminal 20 may assume the section $[t_{y-B+1}^{SL}, n-T_{proc,0}]$ illustrated in FIG. 18 as the sensing window #2. The slot t_{y-B+1}^{SL} corresponds to a slot that is positioned B slots before the first slot t_y^{SL} . The parameter B may be a value associated with the maximum number of slots that can be specified based on the resource allocation field, and may be, for example, 32 slots. Note that some of the slots in the section $[t_{y-B+1}^{SL}, n-T_{proc,0}]$ need not be included in the sensing window #2. Although the determination of the sensing window #2 is made based on the resource allocation field, the sensing operation using the SCI received in the sensing window #2, may be performed based on the resource reservation field or may be performed based on the resource allocation field.

The above-described parameter B may be configured or predefined and may be a parameter indicated to the terminal 20 from the base station 10.

Note that with respect to step S12 and step S13, any one of them may be executed first or both may be executed in parallel.

In step S14, the terminal 20 performs sensing in the sensing window #1 and the sensing window #2. The terminal 20 may detect, for example, that another terminal 20 is performing transmission with the priority p_{RX} in the sensing window #1 and the sensing window #2. If the SCI is detected in the sensing window #1 or in the sensing window #2 and

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RSRP exceeds the threshold $Th_{pTX,pRX}$, a resource in the resource selection window corresponding to the SCI is excluded. If the SCI is detected in the sensing window #1 or the sensing window #2 and the RSRP is less than the threshold $Th_{pTX,pRX}$, a resource in the resource selection window corresponding to the SCI is not excluded. The threshold $Th_{pTX,pRX}$ may be configured or defined for each resource in the sensing window #1 and the sensing window #2, based on, for example, the priority p_{RX} and the priority p_{RX} .

In step S15, each resource in the resource selection window is identified based on the sensing result in step S14, and a set S_A of resource candidates is determined. For example, with respect to the resource selection window of Y slots from t_y^{SL} to t_{y+Y}^{SL} illustrated in FIG. 18, a resource occupied by another UE as illustrated in FIG. 18 is identified, and the resources, from which the identified resource has been excluded, become the available resource candidates. When the set of available resource candidates is S_A , if S_A is less than 20% of the resource selection window, the threshold $Th_{pTX,pRX}$ configured for each resource in the sensing window may be raised by 3 dB to perform resource identification again. That is, by raising the threshold $Th_{pTX,pRX}$ and performing resource identification again, the resources, which are not excluded because RSRP is below the threshold, may be increased, and the set S_A of resource candidates may become 20% or more of the resource selection window. If S_A is less than 20% of the resource selection window, the operation of raising the threshold $Th_{pTX,pRX}$ configured for each resource in the sensing window by 3 dB to perform resource identification again, may be repeated.

In step S16, the terminal 20 selects a resource from the set S_A of resource candidates. For example, in the terminal 20, the S_A may be reported from the lower layer to the higher layer, and random resource selection from the resources included in the S_A , may be performed at the higher layer. The terminal 20 may perform sidelink transmission by using the selected resource.

By performing the sensing and resource selection illustrated in FIGS. 17 and 18, power saving sensing can be achieved in which aperiodic traffic is taken into account.

Here, the number of resources that can be reserved by the resource allocation field of a SCI may be 1. For example, the number of resources may be defined by counting, as a unit, a resource corresponding to one slot and one or more subchannels. For example, a parameter Nmax, which indicates the number of resources that can be reserved by the resource allocation field, may be set to 1, and reservation of a plurality of resources may be performed only by the resource reservation field. Further, for the resource reservation field, sensing may be performed by the sensing window #1 indicated in step S13 of FIG. 17. By this sensing, a sensing mechanism that takes energy saving into account can be the same as that of LTE.

FIG. 19 is a flowchart illustrating an example (2) of sensing according to an embodiment of the present invention. FIG. 20 is a diagram illustrating an example (2) of sensing according to an embodiment of the present invention. The operation of the pre-emption and/or re-evaluation for the resource of the slot m (also referred to as the "resource m"), which has been selected as illustrated in FIG. 20 or is reserved by the transmitted SCI, may be changed. The state after the resource m is selected in FIG. 18, is illustrated in FIG. 20.

Pre-emption is an operation in which the terminal 20A triggers the re-selection of resources when the resource

already selected by the terminal 20A overlaps with the higher priority resource reservation by another terminal 20B and the SL-RSRP measurement value of the high priority resource reservation is greater than the SL-RSRP threshold associated with the resource.

The reevaluation is an operation in which, the terminal 20 has already selected a resource, and before transmitting a SCI for reservation in the selected resource, the terminal 20 identifies resource candidates in the resource selection window and selects a resource from the identified resource candidates. For the resource of slot m , re-evaluation does not need to be triggered at or after the slot $m-T_3$. That is, T_3 corresponds to the processing time required for resource reselection at the terminal 20.

For step S21 through step S23, the same processes as step S14 through step S16 illustrated in FIG. 17 may be performed, or other sensing methods may be performed. In step S23, the terminal 20 selects a resource m from a set S_A of resource candidates. After selecting resource m from the set S_A of resource candidates in step S23, the terminal 20 may use resource m without executing step S24 or may execute step S24.

In step S24, the terminal 20 performs pre-emption and/or re-evaluation for the selected resource m . For the resource selection operation pertaining to the sensing window and resource selection window in step S24, the operation as illustrated in FIG. 18 may be applied or the operation as illustrated in FIG. 14 may be applied. Further, when the operation illustrated in FIG. 18 is applied, parameters A , B , and k may be configured or predefined independently for pre-emption and/or reevaluation, or common parameters may be configured or predefined for pre-emption and/or reevaluation and for the regular resource selection.

In step S24, the terminal 20 may execute additional resource exclusion with respect to the S_A at the time of selection of the resource m . For example, as illustrated in FIG. 20, the terminal 20 may execute additional resource exclusion based on the sensing result of the sensing window #3 with the section $[n-T_{proc,0}, m-T_4)$ as the sensing window #3. Here, T_4 may be, for example, $T_3+T_{proc,0}$ or may be the time required for processing related to resource reselection. The terminal 20 may execute the reselection of resources from the S_A in which the resource identification result has been updated, based on the sensing result of the sensing window #3 (step S25).

The re-evaluation may be performed only by receiving the SCI reserving the resource m from another terminal 20 and determining whether the SL-RSRP value of the SCI exceeds a predetermined threshold. If the SL-RSRP value of the SCI exceeds the predetermined threshold, the terminal 20 performs resource reselection. The terminal 20 may perform sidelink transmission by using the reselected resource.

By the sensing and resource selection illustrated in FIGS. 19 and 20, pre-emption and/or re-evaluation that takes energy saving into account, can be defined.

According to the embodiments described above, the terminal 20 may implement power saving sensing that takes aperiodic traffic into account. The terminal 20 may also perform pre-emption and/or re-evaluation considering power saving. Further, the terminal 20 can simplify the system by making the sensing mechanism in consideration of power saving the same as that of the LTE.

That is, in direct terminal-to-terminal communication, in the case where the resource used by the terminal is autonomously selected, the sensing targets can be reduced.

(Apparatus Configuration)

Next, example functional arrangements of the base station 10 and the terminal 20 that perform operations and actions as stated above are described. The base station 10 and the terminal 20 include functions of implementing the above-stated embodiments. Note that the base station 10 and the terminal 20 each may have only a portion of the functions of the embodiments.

<Base Station 10>

FIG. 21 illustrates an example functional arrangement of the base station 10. As shown in FIG. 21, the base station 10 includes a transmission unit 110, a reception unit 120, a configuration unit 130 and a control unit 140. The functional arrangement shown in FIG. 21 is only one example. The functional separation and the names of the functional units may be arbitrary as long as operations according to the present embodiment can be achieved.

The transmission unit 110 includes a function of generating a signal for transmission to the side of the terminal 20 and wirelessly transmitting the signal. The reception unit 120 includes a function of receiving various signals transmitted from the terminal 20 and acquiring information for upper layers from the received signals, for example. Also, the transmission unit 110 includes a function of transmitting an NR-PSS, an NR-SSS, an NR-PBCH, a DL/UL control signal, a DL reference signal or the like to the terminal 20.

The configuration unit 130 stores preconfigured configurations and various configurations for transmission to the terminal 20 in a memory device and reads them from the memory device as needed. Contents of the configurations may be information associated with configurations of D2D information or the like, for example.

The control unit 140 performs operations associated with configurations for the terminal 20 to perform D2D communication as stated in conjunction with the embodiments. Also, the control unit 140 transmits a scheduling for D2D communication and DL communication to the terminal 20 via the transmission unit 110. Also, the control unit 140 receives information associated with an HARQ acknowledgement for D2D communication and DL communication from the terminal 20 via the reception unit 120. The functional portions of the control unit 140 related to signal transmission may be included in the transmission unit 110, and the functional portions of the control unit 140 related to signal reception may be included in the reception unit 120.

<Terminal 20>

FIG. 22 is a diagram illustrating one example functional arrangement of the terminal 20 according to an embodiment of the present invention. As illustrated in FIG. 22, the terminal 20 has a transmission unit 210, a reception unit 220, a configuration unit 230 and a control unit 240. The functional arrangement shown in FIG. 22 is only one example. The functional separation and the names of the functional units may be arbitrary as long as operations according to the present embodiment can be achieved.

The transmission unit 210 generates a transmission signal from transmission data and wirelessly transmits the transmission signal. The reception unit 220 wirelessly receives various signals and acquires signals for upper layers from the received physical layer signals. Also, the reception unit 220 has a function of receiving an NR-PSS, an NR-SSS, an NR-PBCH, a DL/UL/SL control signal or a reference signal and so on transmitted from the base station 10. Also, for example, as D2D communication, the transmission unit 210 transmits a PSCCH (Physical Sidelink Control Channel), a PSSCH (Physical Sidelink Shared Channel), a PSDCH (Physical Sidelink Discovery Channel), a PSBCH (Physical

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Sidelink Broadcast Channel) or the like to other terminals **20**, and the reception unit **220** receives the PSSCH, the PSSCH, the PSDCH, the PSBCH or the like from other terminals **20**.

The configuration unit **230** stores various configurations received at the reception unit **220** from the base station **10** or the terminal **20** in a memory device and reads them from the memory device as needed. Also, the configuration unit **230** stores preconfigured configurations. Contents of the configurations may be information associated with configuration of D2D communication and so on, for example.

The control unit **240** controls D2D communication to other terminals **20** as stated above in conjunction with the embodiments. Also, the control unit **240** performs operations associated with an HARQ for D2D communication and DL communication. Also, the control unit **240** transmits information associated with an HARQ acknowledgement for D2D communication and DL communication to other terminal **20** scheduled from the base station **10** to the base station **10**. Also, the control unit **240** may perform scheduling for D2D communication for other terminals **20**. Also, the control unit **240** may autonomously select a resource for use in D2D communication from a resource selection window, based on a result of sensing. Also, the control unit **240** performs processing related to MCS in transmission and reception of D2D communication. The functional portion of the control unit **240** regarding signal transmission may be included in the transmission unit **210**, and the functional portion of the control unit **240** regarding signal reception may be included in the reception unit **220**.

(Hardware Arrangement)

The block diagrams (FIGS. **21** and **22**) used in describing the above embodiments show blocks of functional units. These functional blocks (components) are implemented by any combination of at least one of hardware and software. In addition, the implementation method of each function block is not particularly limited. That is, each functional block may be implemented using a single device that is physically or logically combined, or may be implemented by directly or indirectly connecting two or more devices that are physically or logically separated (e.g., using wire, radio, etc.) and using these multiple devices. The functional block may be implemented by combining software with the above-described one device or the above-described plurality of devices.

Functions include, but are not limited to, judgment, decision, determination, computation, calculation, processing, derivation, research, search, verification, reception, transmission, output, access, resolution, choice, selection, establishment, comparison, assumption, expectation, deeming, broadcasting, notifying, communicating, forwarding, configuring, reconfiguring, allocating, mapping, assigning, etc. For example, a functional block (component) that functions to transmit is called a transmitting unit or a transmitter. In either case, as described above, the implementation method is not particularly limited.

For example, each of the base station **10**, the terminal **20** and so on according to one embodiment of the present invention may function as a computer performing operations for a radio communication method according to the present disclosure. FIG. **23** is a diagram illustrating an example of a hardware configuration of the base station **10** and the terminal **20** according to one embodiment of the present disclosure. The base station **10** and the terminal **20** as stated above may be physically configured as a computer device including a processor **1001**, a memory **1002**, a storage **1003**,

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a communication device **1004**, an input device **1005**, an output device **1006**, a bus **1007**, etc.

In the following description, the term “device” can be read as a circuit, a device, a unit, etc. The hardware configuration of the base station **10** and the terminal **20** may be configured to include one or more of the respective devices shown in the figure, or may be configured without some devices.

Each function of the base station **10** and the terminal **20** may be implemented by loading predetermined software (program) on hardware, such as the processor **1001** and the memory **1002**, so that the processor **1001** performs computation and controls communication by the communication device **1004**, and at least one of reading and writing of data in the memory **1002** and the storage **1003**.

The processor **1001**, for example, operates an operating system to control the entire computer. The processor **1001** may be configured with a central processing unit (CPU: Central Processing Unit) including an interface with a peripheral device, a control device, a processing device, a register, etc. For example, the above-stated control units **140** and **240** or the like may be implemented with the processor **1001**.

Additionally, the processor **1001** reads a program (program code), a software module, data, etc., from at least one of the storage **1003** and the communication device **1004** to the memory **1002**, and executes various processes according to these. As the program, a program is used which causes a computer to execute at least a part of the operations described in the above-described embodiment. For example, the control unit **140** of the base station **10** shown in FIG. **21** may be implemented by a control program that is stored in the memory **1002** and that is operated by the processor **1001**. Also, for example, the control unit **240** of the terminal **20** shown in FIG. **22** may be implemented by a control program that is stored in the memory **1002** and that is operated by the processor **1001**. While the various processes described above are described as being executed in one processor **1001**, they may be executed simultaneously or sequentially by two or more processors **1001**. The processor **1001** may be implemented by one or more chips. The program may be transmitted from a network via a telecommunications line.

The memory **1002** is a computer readable storage medium, and, for example, the memory **1002** may be formed of at least one of a ROM (Read Only Memory), an EPROM (Erasable Programmable ROM), an EEPROM (Electrically Erasable Programmable ROM), a RAM (Random Access Memory), etc. The memory **1002** may be referred to as a register, a cache, a main memory (main storage device), etc. The memory **1002** may store a program (program code), a software module, etc., which can be executed for implementing the radio communication method according to one embodiment of the present disclosure.

The storage **1003** is a computer readable storage medium and may be formed of, for example, at least one of an optical disk, such as a CD-ROM (Compact Disc ROM), a hard disk drive, a flexible disk, an optical magnetic disk (e.g., a compact disk, a digital versatile disk, a Blu-ray (registered trademark) disk, a smart card, a flash memory (e.g., a card, a stick, a key drive), a floppy (registered trademark) disk, a magnetic strip, etc. The storage **1003** may be referred to as an auxiliary storage device. The above-described storage medium may be, for example, a database including at least one of the memory **1002** and the storage **1003**, a server, or any other suitable medium.

The communication device **1004** is hardware (transmitting and receiving device) for performing communication

between computers through at least one of a wired network and a wireless network, and is also referred to, for example, as a network device, a network controller, a network card, a communication module, etc. The communication device **1004** may be configured to include, for example, a high frequency switch, a duplexer, a filter, a frequency synthesizer, etc., to implement at least one of frequency division duplex (FDD: Frequency Division Duplex) and time division duplex (TDD: Time Division Duplex). For example, a transceiver antenna, an amplification unit, a transceiver unit, a channel interface or the like may be implemented with the communication device **1004**. The transceiver unit may have an implementation with the transmission unit and the reception unit that are physically or logically separated.

The input device **1005** is an input device (e.g., a keyboard, a mouse, a microphone, a switch, a button, a sensor, etc.) that receives an external input. The output device **1006** is an output device (e.g., a display, speaker, LED lamp, etc.) that performs output toward outside. The input device **1005** and the output device **1006** may be configured to be integrated (e.g., a touch panel).

Each device, such as processor **1001** and memory **1002**, is also connected by the bus **1007** for communicating information. The bus **1007** may be formed of a single bus or may be formed of different buses between devices.

Also, the base station **10** and the terminal **20** may include hardware, such as a microprocessor, a digital signal processor (DSP: Digital Signal Processor), an ASIC (Application Specific Integrated Circuit), a PLD (Programmable Logic Device), and a FPGA (Field Programmable Gate Array), which may implement some or all of each functional block. For example, the processor **1001** may be implemented using at least one of these hardware components.

Overview of Embodiment

As described above, according to the embodiment of the present invention, there is provided a terminal including a receiving unit configured to perform sensing of resource reservation information and resource allocation information transmitted from another terminal by using a sensing window; a control unit configured to determine an available resource set, by identifying a predetermined resource in a resource selection window, based on the resource reservation information and the resource allocation information; and a transmitting unit configured to execute transmission to the another terminal by using a resource selected from the resource set.

With the above configuration, the terminal can implement power-saving sensing that takes aperiodic traffic into account. Further, the terminal **20** can perform pre-emption and/or reevaluation considering power saving. That is, in direct terminal-to-terminal communication, when the resources used by the terminal are autonomously selected, the sensing targets can be reduced.

The receiving unit may perform sensing in a first sensing window and a second sensing window, the first sensing window is determined based on the resource reservation information, and the second sensing window is determined based on the resource allocation information. With this configuration, the terminal **20** can implement power saving sensing that takes aperiodic traffic into account.

The receiving unit may perform the sensing by assuming that all of or a part of slots, which are enabled to reserve a resource in a predetermined section according to the resource reservation information, are a first sensing window, and may perform the sensing by assuming that all of or a part

of slots, which are enabled to reserve a resource in the predetermined section according to the resource allocation information, are a second sensing window. With this configuration, the terminal **20** can implement power saving sensing that takes aperiodic traffic into account.

The control unit may perform additional resource exclusion with respect to the resource set, and the transmitting unit may reselect a resource from the resource set for which the additional resource exclusion has been performed. With this configuration, the terminal **20** can perform pre-emption and/or reevaluation that takes power saving into account.

The receiving unit may perform sensing for the additional resource exclusion in a third sensing window that is determined based on a time required for a process of the resource reselection. With this configuration, the terminal **20** can perform pre-emption and/or reevaluation that takes power saving into account.

Further, according to the embodiment of the present invention, there is provided a communication method executed by a terminal, the communication method including performing sensing of resource reservation information and resource allocation information transmitted from another terminal by using a sensing window; determining an available resource set, by identifying a predetermined resource in a resource selection window, based on the resource reservation information and the resource allocation information; and executing transmission to the another terminal by using a resource selected from the resource set.

With the above configuration, the terminal **20** can implement power-saving sensing that takes aperiodic traffic into account. Further, the terminal **20** can perform pre-emption and/or reevaluation considering power saving. That is, in direct terminal-to-terminal communication, when the resources used by the terminal are autonomously selected, the sensing targets can be reduced.

Supplemental Embodiments

The embodiment of the present invention has been described above, but the disclosed invention is not limited to the above embodiment, and those skilled in the art would understand that various modified examples, revised examples, alternative examples, substitution examples, and the like can be made. In order to facilitate understanding of the present invention, specific numerical value examples are used for explanation, but the numerical values are merely examples, and any suitable values may be used unless otherwise stated. Classifications of items in the above description are not essential to the present invention, contents described in two or more items may be used in combination if necessary, and contents described in an item may be applied to contents described in another item (unless a contradiction arises). The boundaries between the functional units or the processing units in the functional block diagrams do not necessarily correspond to the boundaries of physical components. Operations of a plurality of functional units may be physically implemented by a single component and an operation of a single functional unit may be physically implemented by a plurality of components. Concerning the processing procedures described above in the embodiment, the orders of steps may be changed unless a contradiction arises. For the sake of convenience for describing the processing, the base station **10** and the terminal **20** have been described with the use of the functional block diagrams, but these apparatuses may be implemented by hardware, software, or a combination thereof. Each of software functioning with a processor of the base station apparatus **10**

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according to the embodiment of the present invention and software functioning with a processor of the user equipment **20** according to the embodiment of the present invention may be stored in a random access memory (RAM), a flash memory, a read-only memory (ROM), an EPROM, an EEPROM, a register, a hard disk (HDD), a removable disk, a CD-ROM, a database, a server, or any suitable recording media.

Also, the notification of information is not limited to the aspect or embodiment described in the present disclosure, but may be performed by other methods. For example, the notification of information may be performed by physical layer signaling (for example, DCI (Downlink Control Information), UCI (Uplink Control Information)), higher layer signaling (for example, RRC (Radio Resource Control) signaling, MAC (Medium Access Control) signaling, broadcast information (a MIB (Master Information Block) and a SIB (System Information Block)), other signals, or combinations thereof. The RRC signaling may be also be referred to as an RRC message and may be, for example, an RRC connection setup message, an RRC connection reconfiguration message, or the like.

Each aspect and embodiment described in the present disclosure may be applied to at least one of a system that uses a suitable system such as LTE (Long Term Evolution), LTE-A (LTE-Advanced), SUPER 3G, IMT-Advanced, 4G (4th generation mobile communication system), 5G (5th generation mobile communication system), FRA (Future Radio Access), NR (New Radio), W-CDMA (registered trademark), GSM (registered trademark), CDMA2000, UMB (Ultra Mobile Broadband), IEEE 802.11 (Wi-Fi (registered trademark)), IEEE 802.16 (WiMAX (registered trademark)), IEEE 802.20, UWB (Ultra-WideBand), or Bluetooth (registered trademark), and a next-generation system expanded on the basis thereof. Also, a plurality of systems may be combined and applied (for example, a combination of at least one of LTE and LTE-A with 5G, and the like).

In the operation procedures, sequences, flowcharts, and the like according to each aspect and embodiment described in the present disclosure, the orders of steps may be changed unless a contradiction arises. For example, in the methods described in the present disclosure, elements of various steps are illustrated by using an example order and the methods are not limited to the specific orders presented.

The specific operations performed by the base station **10** described in the present disclosure may in some cases be performed by an upper node. It is clear that, in a network that includes one or more network nodes including the base station **10**, various operations performed for communication with the terminal **20** can be performed by at least one of the base station **10** and another network node other than the base station **10** (for example, a MME, a S-GW, or the like may be mentioned, but not limited thereto). In the above, the description has been made for the case where another network node other than the base station **10** is a single node as an example. However, the other network node may be a combination of a plurality of other network nodes (for example, an MME and a S-GW).

Information, signals, or the like described in the present disclosure may be output from an upper layer (or a lower layer) to a lower layer (or an upper layer). Information, signals, or the like described in the present disclosure may be input and output via a plurality of network nodes.

Information or the like that has been input or output may be stored at a predetermined location (for example, a memory) and may be managed with the use of a management table. Information or the like that is input or output can

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be overwritten, updated, or appended. Information or the like that has been output may be deleted. Information or the like that has been input may be transmitted to another apparatus.

In the present disclosure, determination may be made with the use of a value expressed by one bit (0 or 1), may be made with the use of a Boolean value (true or false), and may be made through a comparison of numerical values (for example, a comparison with a predetermined value).

Regardless of whether software is referred to as software, firmware, middleware, microcode, a hardware description language, or another name, software should be interpreted broadly to mean instructions, instruction sets, codes, code segments, program codes, a program, a sub-program, a software module, an application, a software application, a software package, a routine, a subroutine, an object, an executable file, an execution thread, a procedure, a function, and the like.

Also, software, instructions, information, or the like may be transmitted and received through transmission media. For example, in a case where software is transmitted from a website, a server or another remote source through at least one of wired technology (such as a coaxial cable, an optical-fiber cable, a twisted pair, or a digital subscriber line (DSL)) and radio technology (such as infrared or microwaves), at least one of the wired technology and the radio technology is included in the definition of a transmission medium.

Information, signals, and the like described in the present disclosure may be expressed with the use of any one of various different technologies. For example, data, instructions, commands, information, signals, bits, symbols, chips, and the like mentioned herein throughout the above explanation may be expressed by voltages, currents, electromagnetic waves, magnetic fields or magnetic particles, optical fields or photons, or any combinations thereof.

The terms described in the present disclosure and the terms necessary for understanding the present disclosure may be replaced with terms having the same or similar meanings. For example, at least one of a channel and a symbol may be a signal (signaling). A signal may be a message. A component carrier (CC) may be referred to as a carrier frequency, a cell, a frequency carrier, or the like.

The terms “system” and “network” used in the present disclosure are used interchangeably.

Also, information, parameters, and the like described in the present disclosure may be expressed by absolute values, may be expressed by relative values with respect to predetermined values, and may be expressed by corresponding different information. For example, radio resources may be indicated by indices.

The above-described names used for the parameters are not restrictive in any respect. In addition, formulas or the like using these parameters may be different from those explicitly disclosed in the present disclosure. Various channels (for example, a PUCCH, a PDCCCH, and the like) and information elements can be identified by any suitable names, and therefore, various names given to these various channels and information elements are not restrictive in any respect.

In the present disclosure, terms such as “base station (BS)”, “radio base station”, “base station apparatus”, “fixed station”, “NodeB”, “eNodeB (eNB)”, “gNodeB (gNB)”, “access point”, “transmission point”, “reception point”, “transmission/reception point”, “cell”, “sector”, “cell group”, “carrier”, “component carrier”, and the like may be

used interchangeably. A base station may be referred to as a macro-cell, a small cell, a femtocell, a pico-cell, or the like.

A base station can accommodate one or a plurality of (for example, three) cells. In a case where a base station accommodates a plurality of cells, the whole coverage area of the base station can be divided into a plurality of smaller areas. For each smaller area, a base station subsystem (for example, an indoor miniature base station RRH (Remote Radio Head)) can provide a communication service. The term “cell” or “sector” denotes all or a part of the coverage area of at least one of a base station and a base station subsystem that provides communication services in the coverage.

In the present disclosure, terms such as “mobile station (MS)”, “user terminal”, “user equipment (UE)”, and “terminal” may be used interchangeably.

By the person skilled in the art, a mobile station may be referred to as any one of a subscriber station, a mobile unit, a subscriber unit, a wireless unit, a remote unit, a mobile device, a wireless device, a wireless communication device, a remote device, a mobile subscriber station, an access terminal, a mobile terminal, a wireless terminal, a remote terminal, a handset, a user agent, a mobile client, a client, and other suitable terms.

At least one of a base station and a mobile station may be referred to as a transmitting apparatus, a receiving apparatus, a communication device, or the like. At least one of a base station and a mobile station may be an apparatus mounted on a mobile body, or may be a mobile body itself, or the like. A mobile body may be a transporting device (e.g., a vehicle, an airplane, and the like), an unmanned mobile (e.g., a drone, an automated vehicle, and the like), or a robot (of a manned or unmanned type). It is noted that at least one of a base station and a mobile station includes an apparatus that does not necessarily move during a communication operation. For example, at least one of a base station and a mobile station may be an IoT (Internet of Things) device such as a sensor.

In addition, a base station according to the present disclosure may be read as a user terminal. For example, each aspect or embodiment of the present disclosure may be applied to a configuration in which communication between a base station and a user terminal is replaced by communication between a plurality of user equipments 20 (that may be called D2D (Device-to-Device), V2X (Vehicle-to-Everything), or the like). In this case, a user equipment 20 may have above-described functions of the base station apparatus 10. In this regard, a word such as “up” or “down” may be replaced with a word corresponding to communication between terminals (for example, “side”). For example, an uplink channel, a downlink channel, or the like may be replaced with a side channel.

Similarly, a user terminal according to the present disclosure may be read as a base station. In this case, a base station may have above-described functions of the user terminal.

The term “determining” used herein may mean various operations. For example, judging, calculating, computing, processing, deriving, investigating, looking up, searching, inquiring (for example, looking up a table, a database, or another data structure), ascertaining, or the like may be deemed as making determination. Also, receiving (for example, receiving information), transmitting (for example, transmitting information), inputting, outputting, or accessing (for example, accessing data in a memory), or the like may be deemed as making determination. Also, resolving, selecting, choosing, establishing, comparing, or the like may be deemed as making determination. That is, doing a certain

operation may be deemed as making determination. “Determining” may be read as “assuming”, “expecting”, “considering”, or the like.

Each of the terms “connected” and “coupled” and any variations thereof mean any connection or coupling among two or more elements directly or indirectly and can mean that one or a plurality of intermediate elements are inserted among two or more elements that are “connected” or “coupled” together. Coupling or connecting among elements may be physical one, may be logical one, and may be a combination thereof. For example, “connecting” may be read as “accessing”. In a case where the terms “connected” and “coupled” and any variations thereof are used in the present disclosure, it may be considered that two elements are “connected” or “coupled” together with the use of at least one type of a medium from among one or a plurality of wires, cables, and printed conductive traces, and in addition, as some non-limiting and non-inclusive examples, it may be considered that two elements are “connected” or “coupled” together with the use of electromagnetic energy such as electromagnetic energy having a wavelength of the radio frequency range, the microwave range, or the light range (including both of the visible light range and the invisible light range).

A reference signal can be abbreviated as an RS (Reference Signal). A reference signal may be referred to as a pilot depending on an applied standard.

A term “based on” used in the present disclosure does not mean “based on only” unless otherwise specifically noted. In other words, a term “base on” means both “based on only” and “based on at least”.

Any references to elements denoted by a name including terms such as “first” or “second” used in the present disclosure do not generally limit the amount or the order of these elements. These terms can be used in the present disclosure as a convenient method for distinguishing one or a plurality of elements. Therefore, references to first and second elements do not mean that only the two elements can be employed or that the first element should be, in some way, prior to the second element.

“Means” in each of the above-described apparatuses may be replaced with “unit”, “circuit”, “device”, or the like.

In a case where any one of “include”, “including”, and variations thereof is used in the present disclosure, each of these terms is intended to be inclusive in the same way as the term “comprising”. Further, the term “or” used in the present disclosure is intended to be not exclusive-or.

A radio frame may include, in terms of time domain, one or a plurality of frames. Each of one or a plurality of frames may be referred to as a subframe in terms of time domain. A subframe may include, in terms of time domain, one or a plurality of slots. A subframe may have a fixed time length (e.g., 1 ms) independent of numerology.

The numerology may be a communication parameter that is applied to at least one of transmission and reception of a signal or a channel. The numerology may mean, for example, at least one of a subcarrier spacing (SCS), a bandwidth, a symbol length, a cyclic prefix length, a transmission time interval (TTI), the number of symbols per TTI, a radio frame configuration, a specific filtering processing performed by a transceiver in a frequency domain, a specific windowing processing performed by a transceiver in a time domain, and the like.

A slot may include, in terms of time domain, one or a plurality of symbols (OFDM (Orthogonal Frequency Division Multiplexing) symbols, SC-FDMA (Single Carrier Fre-

quency Division Multiplexing) symbols) symbols, or the like). A slot may be a time unit based on the numerology.

A slot may include a plurality of minislots. Each minislot may include one or a plurality of symbols in terms of the time domain. A minislot may also be referred to as a subslot. A minislot may include fewer symbols than a slot. A PDSCH (or PUSCH) transmitted at a time unit greater than a minislot may be referred to as a PDSCH (or PUSCH) mapping type A. A PDSCH (or PUSCH) transmitted using minislots may be referred to as a PDSCH (or PUSCH) mapping type B.

Each of a radio frame, a subframe, a slot, a minislot, and a symbol means a time unit for transmitting a signal. Each of a radio frame, a subframe, a slot, a minislot, and a symbol may be referred to as other names respectively corresponding thereto.

For example, one subframe may be referred to as a transmission time interval (TTI), a plurality of consecutive subframes may be referred to as a TTI, and one slot or one minislot may be referred to as a TTI. That is, at least one of a subframe and a TTI may be a subframe (1 ms) according to the conventional LTE, may have a period shorter than 1 ms (e.g., 1 to 13 symbols), and may have a period longer than 1 ms. Instead of subframes, units expressing a TTI may be referred to as slots, minislots, or the like.

A TTI means, for example, a minimum time unit of scheduling in radio communication. For example, in an LTE system, a base station performs scheduling for each user equipment 20 to allocate, in TTI units, radio resources (such as frequency bandwidths, transmission power, and the like that can be used by each user equipment 20). However, the definition of a TTI is not limited thereto.

A TTI may be a transmission time unit for channel-coded data packets (transport blocks), code blocks, code words, or the like, and may be a unit of processing such as scheduling, link adaptation, or the like. When a TTI is given, an actual time interval (e.g., the number of symbols) to which transport blocks, code blocks, code words, or the like are mapped may be shorter than the given TTI.

In a case where one slot or one minislot is referred to as a TTI, one or a plurality of TTIs (i.e., one or a plurality of slots or one or a plurality of minislots) may be a minimum time unit of scheduling. The number of slots (the number of minislots) included in the minimum time unit of scheduling may be controlled.

A TTI having a time length of 1 ms may be referred to as an ordinary TTI (a TTI according to LTE Rel. 8-12), a normal TTI, a long TTI, an ordinary subframe, a normal subframe, a long subframe, a slot, or the like. A TTI shorter than an ordinary TTI may be referred to as a shortened TTI, a short TTI, a partial or fractional TTI, a shortened subframe, a short subframe, a minislot, a subslot, a slot, or the like.

Note that a long TTI (for example, a normal TTI, a subframe, and the like) may be read as TTI having a time length exceeding 1 ms, and a short TTI (for example, a shortened TTI) may be read as a TTI having a TTI length less than the TTI length of the long TTI and equal to or more than 1 ms.

A resource block (RB) is a resource allocation unit in terms of a time domain and a frequency domain and may include one or a plurality of consecutive subcarriers in terms of frequency domain. The number of subcarriers included in an RB may be the same regardless of the numerology, and, for example, may be 12. The number of subcarriers included in a RB may be determined based on the numerology.

Also, in terms of the time domain, an RB may include one or a plurality of symbols, and may have a length of 1

minislot, 1 subframe, or 1 TTI. Each of 1 TTI, 1 subframe, and the like may include one or a plurality of resource blocks.

One or a plurality of RBs may be referred to as physical resource blocks (PRBs: Physical RBs), a subcarrier group (SCG: Sub-Carrier Group), a resource element group (REG: Resource Element Group), a PRB pair, an RB pair, or the like.

Also, a resource block may include one or a plurality of resource elements (RE: Resource Elements). For example, 1 RE may be a radio resource area of 1 subcarrier and 1 symbol.

A bandwidth part (BWP) (which may be called a partial bandwidth or the like) may mean a subset of consecutive common RBs (common resource blocks) for certain numerology, in any given carrier. A common RB may be identified by a RB index with respect to a common reference point in the carrier. PRBs may be defined by a BWP and may be numbered in the BWP.

A BWP may include a BWP (UL BWP) for UL and a BWP (DL BWP) for DL. For a UE, one or a plurality of BWPs may be set in 1 carrier.

At least one of configured BWPs may be active, and a UE need not assume sending or receiving a predetermined signal or channel outside the active BWP. A “cell”, a “carrier” or the like in the present disclosure may be read as a “BWP”.

The above-described structures of radio frames, subframes, slots, minislots, symbols, and the like are merely examples. For example, the number of subframes included in a radio frame, the number of slots included in a subframe or a radio frame, the number of minislots included in a slot, the number of symbols and the number of RBs included in a slot or a minislot, the number of subcarriers included in an RB, the number of symbols included in a TTI, a symbol length, a cyclic prefix (CP) length, and the like can be variously changed.

Throughout the present disclosure, in a case where an article such as “a”, “an”, or “the” in English is added through a translation, the present disclosure may include a case where a noun following the article is of a plural form.

Throughout the present disclosure, an expression that “A and B are different” may mean that “A and B are different from each other”. Also, this term may mean that “each of A and B is different from C”. Terms such as “separate” and “coupled” may also be interpreted in a manner similar to “different”.

Each aspect or embodiment described in the present disclosure may be solely used, may be used in combination with another embodiment, and may be used in a manner of being switched with another embodiment upon implementation. Notification of predetermined information (for example, notification of “being x”) may be implemented not only explicitly but also implicitly (for example, by not notifying predetermined information).

Note that S_A or S_B in the present disclosure is an example of an available resource set. The Y slots are an example of a resource in a predetermined section in the resource selection window.

Although the present disclosure has been described in detail above, it will be obvious to those skilled in the art that the present disclosure is not limited to the embodiments described herein. The present disclosure can be implemented as modifications and variations without departing from the spirit and scope of the present disclosure as defined by the scope of the claims.

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Therefore, the description of the present disclosure is for the purpose of illustration and does not have any restrictive meaning to the present disclosure.

REFERENCE SIGNS LIST

10 base station
 110 transmitting unit
 120 receiving unit
 130 configuring unit
 140 control unit
 20 terminal
 210 transmitting unit
 220 receiving unit
 230 configuring unit
 240 control unit
 1001 processor
 1002 storage device
 1003 auxiliary storage device
 1004 communication device
 1005 input device
 1006 output device

The invention claimed is:

1. A terminal comprising:

a processor configured to select a specific number of slots in a specific time interval, to determine, based on a specific slot from among the selected specific number of slots, at least one of a periodic first time interval for performing a first partial sensing or a second time interval for performing a second partial sensing, and to determine, based on at least one of a first result of the first partial sensing or a second result of the second partial sensing, a set of resources usable within the selected specific number of slots; and

a transmitter configured to perform transmission to another terminal by using a resource selected from the set of resources,

wherein the first partial sensing is a partial sensing for a periodic traffic, and the processor determines the first time interval based on a slot located an integral multiple of a specific period before the specific slot from among the selected specific number of slots, and

wherein the processor determines the specific period based on a parameter transmitted from a base station to the terminal or a preconfigured parameter.

2. The terminal as claimed in claim 1, wherein the processor determines, based on the specific slot from among the selected specific number of slots, the first

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time interval and the second time interval, and determines the set of resources based on the first result of the first partial sensing and the second result of the second partial sensing.

3. The terminal as claimed in claim 1, wherein the second partial sensing is a partial sensing for an aperiodic traffic, and

the processor determines the second time interval based on a slot located a predetermined number of slots before the specific slot from among the selected specific number of slots.

4. The terminal as claimed in claim 1, wherein the processor determines the specific number of slots based on a parameter transmitted from a base station to the terminal or a preconfigured parameter.

5. The terminal as claimed in claim 1, wherein when the processor performs at least one of pre-emption or re-evaluation for the selected resource, the processor controls execution of at least one of the first partial sensing or the second partial sensing.

6. The terminal as claimed in claim 5, wherein when the processor performs at least one of pre-emption or re-evaluation for the selected resource, the processor controls execution of at least one of the first partial sensing or the second partial sensing.

7. A communication method performed by a terminal, the communication method comprising:

selecting a specific number of slots in a specific time interval, determining, based on a specific slot from among the selected specific number of slots, at least one of a periodic first time interval for performing a first partial sensing or a second time interval for performing a second partial sensing, and determining, based on at least one of a first result of the first partial sensing or a second result of the second partial sensing, a set of resources usable within the selected specific number of slots; and

performing transmission to another terminal by using a resource selected from the set of resources,

wherein the first partial sensing is a partial sensing for a periodic traffic, and the terminal determines the first time interval based on a slot located an integral multiple of a specific period before the specific slot from among the selected specific number of slots, and

wherein the terminal determines the specific period based on a parameter transmitted from a base station to the terminal or a preconfigured parameter.

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